Multiobjective Hyperspectral Feature Selection Based on Discrete Sine Cosine Algorithm

Yuting Wan, Ailong Ma, Yanfei Zhong, *Senior Member, IEEE*, Xin Hu, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Feature selection is an effective way to reduce the data dimensionality of hyperspectral imagery and obtain a better performance in the subsequent applications, such as classification. The ideal approach is to obtain the optimal tradeoff between two criteria for hyperspectral image feature selection: 1) information preservation and 2) redundancy reduction. However, constructing a hyperspectral feature selection model for the above two criteria is difficult due to the complexity of hyperspectral imagery. Although evolutionary multiobjective optimization methods have been recently presented to simultaneously optimize the above criteria, they cannot control the global exploration versus local exploitation capabilities in the search space for the hyperspectral feature selection problem. Thus, in this article, a novel discrete sine cosine algorithm (SCA)-based multiobjective feature selection (MOSCA_FS) approach is proposed for hyperspectral imagery. In the proposed method, a novel and effective framework of multiobjective hyperspectral feature selection is designed. In the framework, the ratio between the Jeffries–Matusita (JM) distance and mutual information (MI) is modeled to minimize the redundancy and maximize the relevance of the selected feature subset. In addition, another measurement—the variance (Var)—is applied for maximizing the information amount. Furthermore, to resolve the discrete hyperspectral feature selection problem, a novel discrete SCA is first proposed, which enhances the selection of the ideal feature subset. The effectiveness and universality of the proposed method was verified by experiments with ten University of California at Irvine (UCI) data sets, five hyperspectral image data sets, and one spectral data set of typical surface features.**

***Index Terms*— Discrete sine cosine algorithm (SCA), feature selection, hyperspectral remote sensing image, multiobjective optimization, typical surface features.**

NOMENCLATURE

NS	Number of selected features.
VSF	V-shaped function.
$T_{\max}$	Maximum number of iterations.
NP	Population size.
G_i	ith solution in the population.
M, N	Column and row number of the hyperspectral image.
x_{jk}^i	Value of the jth column and kth row in the ith selected feature band image.
ip_{ij}	Incremental part in the jth gene position of the ith solution.
sig	Sigmoid function.
${}^*G_i$	Improved solution using the VSF.

Manuscript received May 1, 2019; revised September 18, 2019 and November 11, 2019; accepted November 28, 2019. Date of publication January 6, 2020; date of current version April 22, 2020. This work was supported in part by the National Key Research and Development Program of China under Grant 2017YFB0504202, in part by the National Natural Science Foundation of China under Grant 41622107, Grant 41771385, and Grant 41801267, and in part by the China Postdoctoral Science Foundation under Grant 2017M622522. *(Corresponding authors: Ailong Ma; Yanfei Zhong.)*

The authors are with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430079, China, and also with the Collaborative Innovation Center of Geospatial Technology, Wuhan University, Wuhan 430079, China (e-mail: wanyuting@whu.edu.cn; maailong007@whu.edu.cn; zhongyanfei@whu.edu.cn; whu_huxin@whu.edu.cn; zlp62@whu.edu.cn).

Color versions of one or more of the figures in this article are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TGRS.2019.2958812

I. INTRODUCTION

ALTHOUGH hyperspectral remote sensing images contain hundreds of feature bands with a fine spectral resolution, the large data volume and redundancy of information often lead to unsatisfactory interpretation results. Thus, hyperspectral feature selection methods have been proposed to resolve these concerns. These methods can be regarded as a combination of a search strategy for obtaining a pivotal hyperspectral feature subset, as well as an evaluation measure for scoring the different feature subsets. In general, the main reasons for using a hyperspectral feature selection strategy are: 1) to avoid the "dimensional disaster" problem caused by hyperspectral imagery; 2) to allow the analytical model to more easily interpret the hyperspectral imagery [1]; and 3) to enhance the generalization by reducing the overfitting.

So what kind of feature subset from the original hyperspectral image do we want to obtain? For a hyperspectral image, when we select an unsatisfactory feature subset, the subsequent analysis model will be difficult to interpret, it will be more likely to lead to overfitting, and it will have low prediction accuracy, in applications such as land-cover classification. In contrast, an ideal feature subset will make the analysis model easy to interpret, less likely to result in overfitting, and will usually obtain high prediction accuracy [2]–[5]. From the existing studies of hyperspectral feature selection, the frameworks can be roughly divided into three main categories [6]: 1) wrapper-based methods, such as the sequential search methods [7], [8], including sequential forward selection (SFS), sequential backward selection (SBS), sequential forward floating selection (SFFS), and sequential backward floating selection (SBFS); 2) filter-based methods, such as information theory-based selection [9]–[11] and minimum estimated abundance covariance-based selection [12], [13]; and 3) embedded-based methods, such as the genetic embedded approach [14], [15]. Fig. 1 depicts the differences in the implementation mechanisms of the different hyperspectral feature selection frameworks, and the "performance" denotes the methods that have utilized the classifier or predictor to

0196-2892 © 2020 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

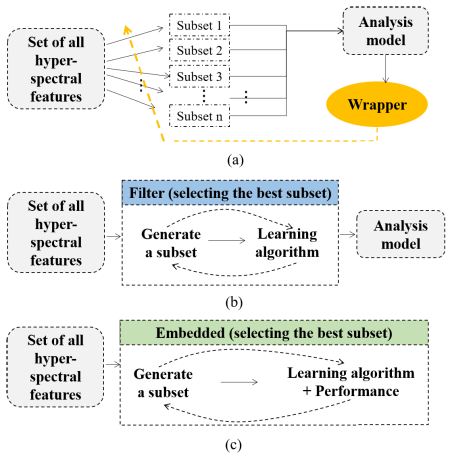


Fig. 1. Frameworks of the three main hyperspectral feature selection approaches. (a) Wrapper-based. (b) Filter-based. (c) Embedded-based.

evaluate the performance of the selected feature subset during the optimization iteration.

In addition, some other efficient hyperspectral feature selection methods have also been proposed, such as the maximum-variance principal component analysis (PCA) methods [16]–[20], 3-D Gaussian–Gabor based method [21], and clustering-based approaches [22]–[26]. However, the optimization mechanisms used in the existing hyperspectral feature selection methods are mostly based on complete search, greedy search, and random search [6], [27]–[29]. Thus, these approaches suffer from a high computational cost and can easily fall into local optima with an unsatisfactory hyperspectral feature subset [30], [31]. Taking these two concerns into consideration, the evolutionary computation (EC)-based hyperspectral feature selection methods can provide ideal solutions and have been the subject of much research interest [32].

The evolutionary single-objective-based hyperspectral feature selection methods were the first EC-based methods to be proposed. The genetic algorithm-based methods were likely the first EC-based methods to be widely utilized in a hyperspectral feature selection framework [33]–[36]. Subsequently, other EC-based methods have also been used in single-objective hyperspectral feature selection, including the particle swarm optimization (PSO) algorithm [12], the differential evolution algorithm [37], [38], the clonal selection algorithm (CSA) [9], [39], [40], artificial bee colony algorithm [41], and the sine cosine algorithm (SCA) [42]. When embedded into the wrapper-based approaches, the EC-based hyperspectral feature selection methods can be utilized to search for the best feature subset, but a predictor is needed to evaluate the performance under the selected subset. In addition, EC-based methods have also been utilized in filter-based methods for hyperspectral feature selection, where different measures are utilized, such as information theory [10], [38], [43]–[46], the minimum estimated abundance covariance [12], [47], consistency indices [36], [38], fuzzy set theory [48], [49], and rough set theory [50]–[52].

However, although many measures have been introduced for hyperspectral feature selection, the use of only a single measure is a poor way to obtain a hyperspectral feature subset with a maximum amount of information, minimum redundancy, and maximum relevance. For example, the single-objective-based ranking methods do not consider the interactions between hyperspectral features, which often result in a redundant feature subset from the hyperspectral image.

Recently, in order to capture the different characteristics of hyperspectral imagery for feature selection, multiple measures have been considered in a unified framework for hyperspectral feature selection. With the development of multiobjective optimization, and due to the powerful search capability of evolutionary algorithms [53], which originate from the imitation of natural evolution, evolutionary multiobjective optimization methods have been successfully applied in many fields, such as communication and network optimization [54], [55], transportation [56], [57], and financial risk management [58], [59]. These methods have also been applied in many applications in remote sensing image processing, including image clustering [60], [61], hyperspectral image sparse unmixing [62], change detection [63], and subpixel mapping [64], [65]. Thus, multiobjective evolutionary algorithm-based hyperspectral feature selection methods have been proposed to resolve the drawbacks of the single-objective-based methods. For example, Zhang *et al.* [43] adopted variants of information entropy and cross entropy as the two objective functions for hyperspectral band selection, which were optimized through the multiobjective immune algorithm; Xia *et al.* [45] proposed the multiobjective unsupervised feature selection algorithm utilizing a redundancy measure and negative epsilon-dominance for fault diagnosis; and Pan *et al.* [66] developed an adaptive-penalty-based boundary intersection (APBI) framework to improve the multiobjective algorithm in hyperspectral band selection. However, a single evolutionary algorithm used in the multiobjective framework cannot optimize complex problems in hyperspectral feature selection, and it will have difficulty in meeting the needs of multiple types of hyperspectral images. Generally speaking, in order to improve the optimization capability, the common approach is to split the optimization process into two phases: 1) global exploration and 2) local exploitation [67]. The two phases can make a good combination of long-distance and short-distance search in the search space to obtain a better feature subset for the hyperspectral feature selection problem.

In order to obtain a better optimization capability, memetic algorithm (MA)-based feature selection approaches have also been proposed for transient stability assessment [49]. However, for the MA-based methods, the tradeoff between the global and local search capabilities cannot be well controlled, and the multiobjective MA-based hyperspectral feature selection methods need to be further developed.

To search for a better feature subset of the original hyperspectral image, a recently proposed population-based optimization algorithm called the SCA [68] has the ability to better control the balance between the global and local search capabilities, and has been successfully applied in airfoil design [68]. For the applications of multiobjective SCA, it is currently used in a novel hybrid forecasting system for wind speed [69] and multiobjective engineering design

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

problems [70]. In the existing research, the SCA is basically a continuous version, and only one type of discrete SCA has been designed for solving the traveling salesman problem [71]. In [71], the original SCA was applied to implement the updating and searching of the solutions for the best tour between cities in the traveling salesman problem. However, due to the need to use 0–1 encoding in the hyperspectral feature selection problem, which is a typical discrete problem, the continuous version is difficult to use directly and effectively.

In this article, to address these concerns, we propose a novel discrete SCA-based multiobjective hyperspectral feature selection (MOSCA_FS) approach. The main contributions of the proposed method are as follows.

1) *A Novel Framework of Multiobjective Hyperspectral Feature Selection*: To obtain a better tradeoff between the global and local search capabilities and to obtain a refined feature subset with the attributes of the maximum amount of information, minimum redundancy, and maximum relevance, a novel discrete SCA is designed and embedded into the novel framework of multiobjective hyperspectral feature selection. In addition, effective objective functions for selecting the ideal feature subset are also constructed in the framework. In order to achieve the retention of critical information and the removal of redundant information, an effective measurement—the ratio between the Jeffries–Matusita (JM) distance and the mutual information (MI)—is designed, along with another measurement—the variance (Var)—for multiobjective hyperspectral feature selection.
2) *A Discrete SCA for Better Controlling the Balance Between the Global and Local Search Capabilities*: In order to implement the multiobjective hyperspectral feature selection, a novel discrete SCA is first designed, which uses a VSF to better capture the optimization capability, instead of just judging by a probability value, as in the original sigmoid function. Furthermore, this framework can perform better in balancing the global and local search capabilities in the search space for the hyperspectral feature selection problem.

In order to verify the effectiveness and universality of the proposed method, different types of data sets were tested in the experiments undertaken in this study: ten University of California at Irvine (UCI) data sets, five hyperspectral image data sets, and one spectral data set with more feature bands of typical surface features.

The rest of this article is organized as follows. Section II gives the related background, including multiobjective optimization and the SCA. Section III describes the framework of the proposed MOSCA_FS approach. The experiments and analyses, the computational complexity analysis, and the parameter sensitive analysis are presented in Section IV. Finally, Section V draws our conclusions.

II. Related Research Background

A. Multiobjective Optimization

In order to address the concerns in the existing hyperspectral image feature selection methods, more measures need to be considered in the framework, and thus the multiobjective optimization methods were introduced. Compared to a single-objective optimization problem, a multiobjective optimization problem involves optimizing multiple objective functions simultaneously. Some basic definitions for a multiobjective optimization problem are provided below.

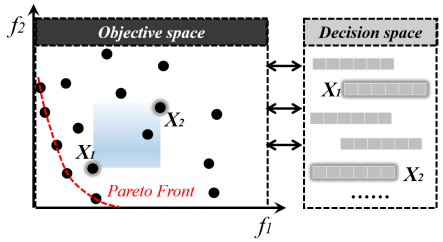


Fig. 2. Corresponding definitions in a multiobjective optimization problem: Pareto domination and the Pareto front.

1) Multiobjective Optimization Problem: We assume that there are m objective functions for the minimum optimization problem, and that they are conflicting with each other, as shown in (1). In addition, the objective functions satisfy constraints (2) and (3)

$$\min f(X) = (f_1(X), f_2(X), \ldots, f_m(X)) \tag{1}$$

$$g_i(X) \geq 0, \quad (i = 1, 2, \ldots, k) \tag{2}$$

$$h_i(X) = 0, \quad (i = 1, 2, \ldots, l). \tag{3}$$

Thus, we seek solutions $X* = (x_1^*, x_2^*, \ldots, x_n^*)$, where they satisfy the conditions in (2) and (3)

$$\forall i = 1, 2, \ldots, m, \quad f_i(X_1) \leq f_i(X_2) \wedge$$
$$\exists j = 1, 2, \ldots, m, \quad f_j(X_1) < f_j(X_2). \tag{4}$$

2) Pareto Domination: For the decision variables X_1 and X_2, when they satisfy the condition in (4), it can be said that X_1 Pareto dominates X_2, as shown in Fig. 2.

3) Pareto Front: The final result of a multiobjective optimization problem is not a single solution, but a Pareto optimal solution set. As shown in Fig. 2, the final solutions form the Pareto optimal solution set, in which they are nondominated with each other. This is also known as the Pareto front.

Hence, the multiobjective optimization methods aim at obtaining a Pareto front with high-quality solutions. The strategies of nondominated sorting and density estimation ranking used in non-dominated sorting genetic algorithm II (NSGA-II) [72] are utilized in MOSCA_FS. In addition, for updating the solutions in a population to obtain the final desired solution, a population-based search method is needed in the evolutionary multiobjective optimization framework. In this article, in order to capture the global and local search capabilities, the recently proposed SCA is introduced.

B. SCA

In order to refine the solutions in a population, the SCA is applied. The SCA was proposed by Mirjalili [68] in 2016 for solving optimization problems, and has since been used in airfoil design. The SCA has a good combination of exploration and exploitation in searching for the solutions. In the initial stage of the SCA, it creates multiple initial random candidate

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

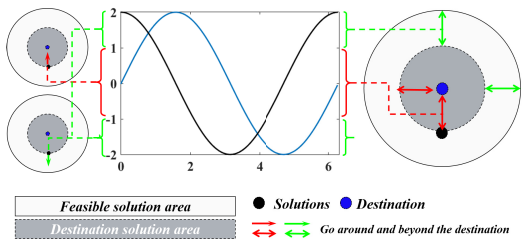


Fig. 3. Search process of the SCA.

solutions to find the promising region of the search space. However, during the development phase, the random solutions gradually change, and the random changes are much smaller than in the exploration phase. The mathematical model used in the SCA is based on the following update function for any search solution:

$$G_i^{t+1} = \begin{cases} G_i^t + r_1 \times \sin(r_2) \times |r_3 P_i^t - G_i^t|, & \text{if } r_4 < 0.5 \\ G_i^t + r_1 \times \cos(r_2) \times |r_3 P_i^t - G_i^t|, & \text{if } r_4 \geq 0.5 \end{cases} \quad (5)$$

$$r_1 = \alpha - t\frac{\alpha}{T_{\max}} \quad (6)$$

where G_i^t is the solution from the current iteration; G_i^{t+1} is the updated solution; P_i^t denotes the destination point (the best solution in a population) in the current iteration; and r_1 is a control parameter that balances the exploration and exploitation in each iteration according to (6). Each of r_2, r_3, and r_4 are random numbers, where $r_2 = \text{rand}(2\pi)$ and $r_3, r_4 = \text{rand}(0, 1)$. t denotes the current iteration, $T_{\max}$ is the maximum number of iterations, and α is a constant.

In addition, there are four main parameters: r_1, r_2, r_3, and r_4. Among them, r_1 determines the next position's area for the search solutions; r_2 dictates how far the movement should be toward or away from the destination point; r_3 is a random weight for the destination point and the search solutions; and r_4 switches equally between the sine and cosine components.

Algorithm 1 Pseudo-Code of the Sine Cosine Algorithm

```
Begin
    Initialize a set of search solution G_i in a population,
    the max number of iterations T_max, the constant α;
    Do (for each iteration)
        Evaluate the solution by the objective function;
        Update the destination point P_i;
        Update the parameters r_1, r_2, r_3, and r_4;
        Update the position of the solution G_i by (5);
    While (t < T_max)
    Return the best solution after the iteration.
End
```

To provide a specific explanation of the optimization process of the SCA, the search process is presented in Fig. 3, and the pseudo-code of the SCA is given in Algorithm 1. This shows how the search process is performed when the range of the sine and cosine functions is changed. It should be noted that the solutions explore the feasible solution region when the values of the sine and cosine functions are in the range (1, 2] and [−2, −1), respectively, and the solutions exploit the destination solution region when the values are in the range [−1, 1].

III. MOSCA_FS

Since the traditional single objective function with a fixed preference cannot achieve a great performance on different and complex data sets, and considering the superior optimization capability of the SCA, a novel multiobjective discrete version is proposed in this article to select the tradeoff feature subset by preserving as much information as possible, while simultaneously reducing the redundancy. The objective functions are then constructed for the feature selection, and the novel discrete SCA is designed for searching for the optimal solution. The main framework of MOSCA_FS is shown in Fig. 4.

A. Effective Multiobjective Model for Feature Selection

For the feature selection problem, the ideal approach is to obtain the optimal tradeoff between the two criteria of: 1) key information preservation and 2) redundancy reduction. Thus, the feature selection problem can be transformed into a multiobjective optimization problem. In order to select the satisfactory and superior feature subset for making the model easy to interpret, less likely to over-fit, and obtain a high prediction accuracy, multiple objective functions are designed for multiobjective feature selection. One of the objective functions is the Var (7), where the image gray Var directly reflects the total discreteness between the gray value of each pixel and the average gray value of the image. It is an important measure used to measure the information quantity of an image [73]. As one of the two objective functions in the proposed method, it can guarantee the maximum amount of image information while removing redundancy. In addition, we want to remove the redundant information in the data set. The JM distance [74] is a common class separability index utilized for feature selection in multiclass problems, where the larger the value of the JM distance (8), the better the data can be separated. Furthermore, the MI (12) is another effective index in information theory that is used to measure the redundant information in a data set, where the smaller the value of the MI, the lower the information redundancy of the data. Thus, the other objective function is the ratio (13) between the JM distance and the MI, which aims to minimize the information redundancy of the data as much as possible by maximizing the value of the ratio. The corresponding formulas are as follows:

$$f_1 = \text{var} = \frac{1}{\text{NS}} \sum_{i=1}^{\text{NS}} \left(\frac{1}{M \times N} \sum_{j=1}^{M} \sum_{k=1}^{N} \left(x_{jk}^i - \overline{x^i} \right)^2 \right) \quad (7)$$

where NS is the number of selected features; $M \times N$ denotes the number of instances; M and N are the numbers of columns and rows in the hyperspectral remote sensing image, respectively; x_{jk}^i is the value of the jth column and kth row in the ith selected feature band image in the hyperspectral remote sensing image; and $\overline{x^i}$ is the average value of the ith selected feature band image in the hyperspectral remote sensing image. Thus, it calculates the sum of squares of each pixel with the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

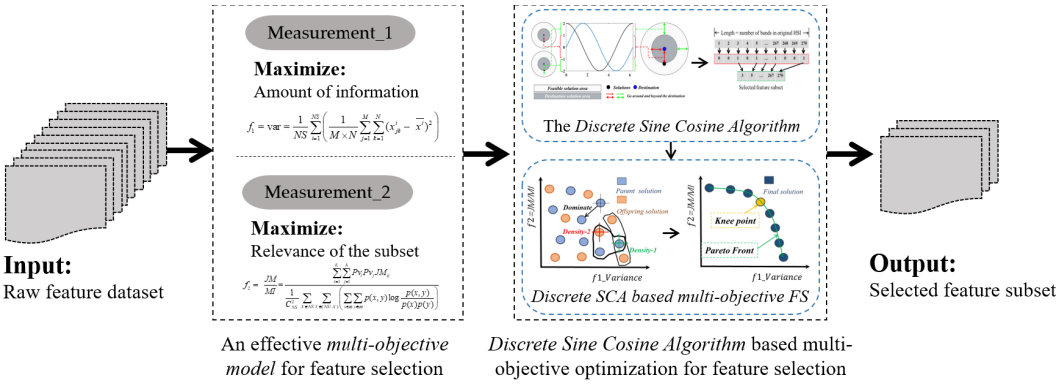


Fig. 4. Main framework of the proposed MOSCA_FS method.

mean value of the pixels in the selected feature band image, and reflects the discreteness of the pixel values, which can measure the amount of image information.

The JM distance is a saturation transform of the Bhattacharyya distance (BD) [75]. It assumes that there are K classes with Gaussian distributions, and the average JM distance of a selected feature subset is calculated through the following equations:

$$\mathrm{JM} = \sum_{i=1}^{K}\sum_{j=1}^{K} \mathrm{Pv}_i \mathrm{Pv}_j \mathrm{JM}_{ij} \tag{8}$$

$$\mathrm{JM}_{ij} = \sqrt{2\left(1 - e^{-\mathrm{BD}_{ij}}\right)} \tag{9}$$

$$\mathrm{BD}_{ij} = \frac{1}{8}(M\upsilon_i - M\upsilon_j)^T \left((\mathrm{cov}_i + \mathrm{cov}_j)/2\right)^{-1}(M\upsilon_i - M\upsilon_j) + \frac{1}{2}\ln\left(\left|\frac{\mathrm{cov}_i + \mathrm{cov}_j}{2}\right| \Big/ \sqrt{|\mathrm{cov}_i|\,|\mathrm{cov}_j|}\right) \tag{10}$$

where K is the number of classes; $P\upsilon_i$ is the *a priori* probability of the ith class; $M\upsilon_i$ is the mean vector of the ith class; and cov_i is the covariance matrix, where $|\bullet|$ represents the determinant.

Next, drawing on the idea of information theory, the MI of the two selected features X and Y is defined as follows:

$$\begin{aligned}\mathrm{MI}(X,Y) &= \sum_{x\in\omega}\sum_{y\in\omega} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}\\ p(x) &= \mathrm{hist}(x)/(M\times N),\ p(x,y)\\ &= \mathrm{hist}(x,y)/(M\times N)\end{aligned} \tag{11}$$

$$\begin{aligned}\mathrm{MI} &= \frac{1}{C_{\mathrm{NS}}^2}\sum_{X\in\mathrm{NS}}\sum_{Y\in(\mathrm{NS}\setminus X)}\mathrm{MI}(X,Y)\\ &= \frac{1}{C_{\mathrm{NS}}^2}\sum_{X\in\mathrm{NS}}\sum_{Y\in(\mathrm{NS}\setminus X)}\\ &\quad\times\left(\sum_{x\in\omega}\sum_{y\in\omega} p(x,y)\log\frac{p(x,y)}{p(x)p(y)}\right).\end{aligned} \tag{12}$$

Thus

$$f_2 = \frac{0}{\mathrm{JM}}\mathrm{MI} \tag{13}$$

where $p(x)$ is the probability mass function; hist (x) is the histogram of X; $p(x,y)$ represents the joint probability distribution function; $\mathrm{hist}(x,y)$ is the joint histogram of X and Y; and C_{NS}^2 denotes the selection of two features from the selected features for calculation of the MI. All the features are combined and calculated in pairs, and the average value is regarded as the final MI value. $X \in \mathrm{NS}$, $Y \in (\mathrm{NS}\setminus X)$ means that X selects a feature from the selected feature subset, and Y selects a feature from the remaining features in the selected feature subset.

B. Discrete SCA

Similar to the PSO algorithm for feature selection [12], the SCA can also be converted into a discrete version. The value of the velocity in PSO is replaced by the incremental part (ip_{ij}) in the SCA, as shown in the following equation:

$$ip_{ij} = \begin{cases} r_1 \times \sin(r_2) \times \left|r_3 P_{ij}^t - G_{ij}^t\right|, & \text{if } r_4 < 0.5 \\ r_1 \times \cos(r_2) \times \left|r_3 P_{ij}^t - G_{ij}^t\right|, & \text{if } r_4 \geq 0.5. \end{cases} \tag{14}$$

In order to express the value of ip as the probability of taking 0 or 1, the sigmoid function (15) is used to make the value of ip map to [0, 1]. The gene is then updated by function (16)

$$\mathrm{sig}(ip_{ij}) = \frac{1}{1+e^{-ip_{ij}}} \tag{15}$$

$$G_{ij}^{t+1} = \begin{cases} 1, & \text{if rand()} \leq \mathrm{sig}(ip_{ij}) \\ 0, & \text{otherwise} \end{cases} \tag{16}$$

where rand() is a random number that is randomly generated from the uniform distribution of interval [0, 1].

However, from (15) and (16), it should be noted that the value of the function denotes the probability when the gene position is taken as "1" or "0." It can be assumed that the probability when the gene position is "1" is $\mathrm{sig}(ip_{ij})$, and then the probability of "0" is $1 - \mathrm{sig}(ip_{ij})$. Thus, the probability of "0" becoming "1" is $\mathrm{sig}(ip_{ij})$, and the probability of "1" becoming "0" is $1 - \mathrm{sig}(ip_{ij})$. The absolute probability of a change in the gene position is expressed as follows:

$$\begin{aligned} p(\Delta) &= \mathrm{sig}(ip_{ij})\left(1 - \mathrm{sig}(ip_{ij})\right) = \mathrm{sig}(ip_{ij}) - \mathrm{sig}(ip_{ij})^2 \\ &= \frac{1}{1+e^{-ip_{ij}}} - \left(\frac{1}{1+e^{-ip_{ij}}}\right)^2. \end{aligned} \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

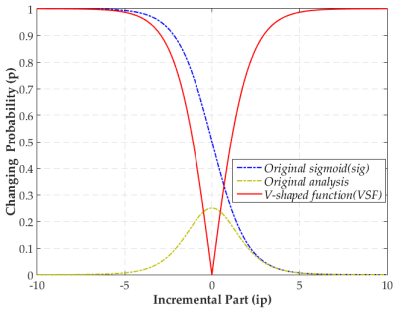


Fig. 5. Analysis and comparison of the VSF and the original sigmoid function, reflecting the relationship between the incremental part and the changing probability of the gene position.

As ip_{ij} denotes the difference between solutions with the destination point, in order to guide the direction of evolution, thus when $ip_{ij} = 0$, the lower probability of change is expected. However, the original analysis for (17) is shown in Fig. 5. Combining the absolute probability and the sigmoid function for the analysis, when $ip_{ij} = 0$, the absolute probability reaches the maximum value, i.e., 0.25. Furthermore, it should be noted that the value of the sigmoid function is 0.5 when the next gene position is closer to the destination point, which is not conducive to the algorithm converging to a global optimal solution. In other words, it cannot provide enough diversity, causing an imbalance between the global exploration and local exploitation [76]. Thus, when we simply use this sigmoid function to discretize the SCA, it is not rigorous enough. The improved version should meet the following conditions: 1) when $ip_{ij} = 0$, the value of the new probability mapping function should be 0 and 2) when $ip_{ij} > 0$ *or* $ip_{ij} < 0$, the probability mapping function should be symmetric about the y-axis. The equation for the new sigmoid function is recorded as a VSF (VSF(ip_{ij})), as presented in (18). The VSF and the original sigmoid function are compared in Fig. 5

$$\mathrm{VSF}(ip_{ij}) = \begin{cases} 1 - \dfrac{2}{1 + e^{-ip_{ij}}} & ip_{ij} \le 0 \\ \dfrac{2}{1 + e^{-ip_{ij}}} - 1 & \text{otherwise.} \end{cases} \tag{18}$$

Equation (18) is then utilized to transform the incremental part into a probability value, by which the gene position of the ith solution is updated by

$${}^{*}G_{ij}^{t+1} = \begin{cases} 0, & \text{if rand}() \le \mathrm{VSF}(ip_{ij}) \text{ and } ip_{ij} \le 0 \\ 1, & \text{if rand}() \le \mathrm{VSF}(ip_{ij}) \text{ and } ip_{ij} > 0 \\ {}^{*}G_{ij}^{t}, & \text{if rand}() > \mathrm{VSF}(ip_{ij}). \end{cases} \tag{19}$$

C. Discrete MOSCA_FS

In this section, after introducing the multiple objective functions for feature selection and the discrete SCA used in the proposed method, the implementation process of MOSCA_FS is described. The pseudocode is presented in Algorithm 2, and descriptions of the individual steps are provided as follows.

Algorithm 2 Pseudo-Code of MOSCA_FS

Input the data and the corresponding label;
Divide *datasetinto a training set and a test set;*
Initialize *a set of the solutions* $\boldsymbol{G}_i$ *in a population* $\boldsymbol{T}_{max}$, α, *and the size of the pre-set feature subset* $\rightarrow$ *num_subset;*
Calculate *the objective functions for each solution; /*****Var*** *and JM/MI */ (7) and (13)*
Ranking: non-dominated and density ranking;
Select *one solution by the knee point based method as the destination point* $\boldsymbol{P}_i$;
  Do *(for each iteration)*
    for *each solution* ***do***
      Update *the parameters* $\boldsymbol{r}_1$, $\boldsymbol{r}_2$, $\boldsymbol{r}_3$, *and* $\boldsymbol{r}_4$;
      Calculate *the incremental part* $\boldsymbol{ip}_{ij}$ *by (14);*
      Calculate *the change probability* $\boldsymbol{p}$ *(17) and* ***VSF*** *(18);*
      Update *the position of the solutions* $\boldsymbol{G}_i$ *by (19) and count the size of the selected feature subset* $\rightarrow$ *num_subset*;*
      if *num_subset** < *num_subset*
        n = num_subset – num_subset;*
        Random {all_feature\(num_subset∪num_subset), n} → Solution*(num_subset*);*
      Elseif *num_subset** > *num_subset*
        Loc_1 = num_subset∩num_subset;*
        Loc_2 = num_subset\Loc_1;*
        Calculate *value of JM/MI {Loc_2(i) with num_subset}*
        Loc_3 = {num_subset – num_subset} →* ***smallest*** *values ofJM/MI; → Solution*(num_subset* =Loc_3) = 0;*
      Else no change;
      End
    End
    Calculate *the objective functions for solution; /****Var*** *and JM/MI */ (7) and (13);*
    Ranking: non-dominated and density ranking;
    Update *the destination point* $\boldsymbol{P}_i$ *through the knee point based method;*
  While $(\boldsymbol{t} < \boldsymbol{T}_{max})$
Output *the best solution after the iteration.*

1) *Step 1 (Input theData Set)*: The data set is first input and is divided into a training set and a test set.
2) *Step 2 (Initialization of the Solutions and the Population)*: According to the preset size of the feature subset (*num_subset*), the number of the "1" gene position in a solution is set as the *num_subset*. The encoding strategy is shown in Fig. 6, where the "0" and "1" gene positions represent the variable values in a solution, and "1" denotes that the corresponding feature is selected while "0" denotes that it is not selected. NP solutions are randomly initialized in a population for subsequent evolutionary operations, which is the set of the solutions.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

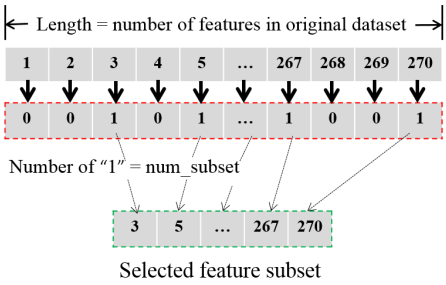


Fig. 6. Encoding strategy of the solutions in a population for feature selection. The "0" and "1" gene positions represent the variable values in a solution, where "1" denotes that the corresponding feature is selected while "0" denotes that it is not selected.

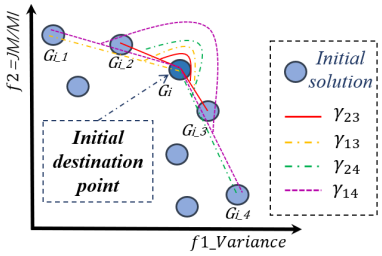


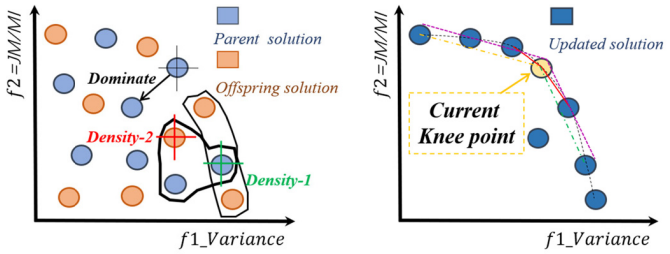
Fig. 7. Knee point-based selection method. After the initialization of the solutions in a population, the corresponding values of the objective functions are calculated, and the tradeoff solution obtained by the knee point-based selection method from the nondominated solutions in the current step is taken as the destination point for the next iteration.

3) *Step 3 (Calculation of the Values of the Objective Functions and the Selection of the Destination Point)*: After the initialization of the population and the solutions, a destination point is needed for the subsequent iteration. Thus, for every solution, the corresponding values of the objective functions are calculated by (7) and (13). The Pareto domination strategy in NSGA-II is used for obtaining superior solutions, and the destination point is then obtained through the knee point-based selection method [77], [78]. The mechanism of the knee point-based selection method is shown in Fig. 7. The four neighbors $(G_{i_1}, G_{i_2}, G_{i_3}, G_{i_4})$ are considered, and the four angles $(\gamma_{23}, \gamma_{13}, \gamma_{24}, \gamma_{14})$ are then calculated. Furthermore, each solution with the maximum angle among $(\gamma_{23}, \gamma_{13}, \gamma_{24}, \gamma_{14})$ is considered as the tradeoff solution. The global tradeoff solution, which is regarded as the initial destination point, can be obtained by comparing all the nondominated solutions.
4) *Step 4 (Evolutionary Operations by Discrete SCA)*: The novel discrete SCA version is then applied to update the solutions in a population by (14), (18), and (19). However, it should be noted that the number of the "1" (num_subset$*$) gene position in the offspring solutions may be more than, less than, or equal to the preset size of the feature subset. In order to address this problem, a simple strategy is conducted.
   a) When num_subset$*$ $<$ num_subset, we first need to calculate the difference for the updated offspring solutions in the new population $n = num_subset - num_subset*$. The union between the two offspring solutions then needs to be calculated as $u = num_subset \cup num_subset*$. The difference set is then obtained by subtracting the union from the entire feature set $D_set = all_feature \backslash u$. Finally, we randomly select the n features in the difference set to add to the solution. This operation can increase the diversity of the population, to a certain extent.
   b) When num_subset$*$ $>$ num_subset, the difference is $n' = num_subset * - num_subset$. The intersection of the selected features is then obtained between the offspring solution and the destination point, i.e., $Loc_1 = num_subset * \cap num_subset$. The features in the intersection are then removed from the offspring solution, i.e., $Loc_2 = num_subset * \backslash Loc_1$, and each feature in *Loc_2* is used to calculate the value of the ratio of JM and MI with the feature subset in the destination point through (13). The n' features with the minimum values of the ratio are then found, and the values of the corresponding gene positions then become "0," which means they are not chosen. These operations can help solutions perform better.
   c) When num_subset$*$ $=$ num_subset, no operations are conducted on the offspring solutions.
5) *Step 5 (Calculation of the Values of the Objective Functions and Updating of the Population and the Destination Point)*: After the evolutionary operation, the size of the population becomes $2 \times$ NP. In order to keep the size of the population a constant, ranking strategies are applied. First, for every solution, the corresponding values of the objective functions are calculated by (7) and (13). The Pareto domination ranking and density ranking are then utilized, as shown in Fig. 8. In addition, the mechanism of the Pareto domination ranking is based on (4), the density ranking is based on (20), and the solutions holding a larger value are selected to enter the next iteration. In addition, after the ranking, the destination point is again selected by the knee point-based method

$$\text{Density}_k = \frac{f_1(G_{:k-1}) - f_1(G_{:k+1})}{f_1^{\max} - f_1^{\min}} + \frac{f_2(G_{:k-1}) - f_2(G_{:k+1})}{f_2^{\max} - f_2^{\min}} \quad (20)$$

where $f_1(\cdot)$ and $f_2(\cdot)$ are the corresponding values of the two objective functions for feature selection, and $f_1^{\min}$, $f_1^{\max}$, $f_2^{\min}$, and $f_2^{\max}$ represent the minimum and maximum values of the two objective functions in a population.
6) *Step 6 (Stopping Condition)*: Steps 4–5 are repeated until the number of iterations is no greater than the maximum value, and the best solution is then obtained through the knee point-based method, which is regarded as the final feature subset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 8. Updating of the population and the destination point. (a) Pareto domination ranking and density ranking. (b) Knee point-based selection.

IV. Experiments

In order to prove the effectiveness of the proposed method—MOSCA_FS—we compared it with several other traditional and state-of-the-art approaches: the traditional maximum-variance PCA method [16]; the SFS method [7]; a composite relevance measure (CRM)-based filter method for feature selection in an automatic target recognition system for underwater mine classification [79], which combines the MI, Shannon's entropy, and the modified relief weight (mRW); the feature selection algorithm called maximum CRM with an SFS scheme (MCRM-SFS); a saliency-based band selection approach inspired by scale selection (SBSS) [80]; and unsupervised band selection by dominant set extraction (DSEBS) [81]. We also compared MOSCA_FS with four evolutionary single-objective feature selection methods, where the two objective functions for feature selection were based on PSO and SCA, respectively. The final comparative method was an evolutionary multiobjective feature selection method, i.e., the multiobjective PSO method (MOPSO). The parameters of the constant α, population size, and maximum number of iterations were set to 2, 30, and 30, respectively.

A. Test Data and Evaluation Criteria

In order to test the universality of the proposed method, three different types of data sets were used: ten UCI data sets, five hyperspectral image data sets, and one spectral data set of typical surface features. In addition, for the performance testing of the selected feature subset, we calculated the classification error rate (21), overall accuracy (OA), and Kappa coefficient (22) using the selected feature subset through the chosen classifier, which was either a support vector machine (SVM) or k-nearest neighbor (KNN) classifier

$$\text{error_rate} = \frac{\text{testing_sample_correct}}{\text{testing_sample_all}} \tag{21}$$

$$\text{OA} = \frac{\sum_{i=1}^{K} \text{Nu}_{ii}}{\text{Nu}}$$

$$\text{Kappa} = \frac{\text{Nu}\sum_{i=1}^{K} \text{Nu}_{ii} - \sum_{i=1}^{K} \text{Nu}_{i+} \times \text{Nu}_{+j}}{\text{Nu}^2 - \sum_{i=1}^{K} \text{Nu}_{i+} \times \text{Nu}_{+j}} \tag{22}$$

where Nu_{ii} is the number of correctly classified samples; Nu_{+j} denotes the total number of samples classified as the jth class in the actual classification results; and Nu_{i+} represents the total number of samples of the ith class in the test samples.

TABLE I
Details of the UCI Data Sets

Dataset	# instances	# features	# classes	Date
1. Balance scale	625	4 (2-4)	3	1994
2. Wine	178	13 (2-10)	3	1991
3. Zoo	101	16 (2-15)	7	1990
4. Waveform v.1	5000	21 (2-20)	3	1988
5. Spambase	4601	57 (2-55)	2	1999
6. Sonar	208	60 (2-60)	2	N/A
7. Musk v.2	6598	166 (2-100)	2	1994
8. Msplice	3175	240 (2-100)	3	N/A
9. Madelon	2600	500 (2-100)	2	2008
10. Isolet5	1560	617 (2-100)	2	1994

The error rate is used to evaluate the performance in the ten UCI data sets, and the OA and Kappa are used for the five hyperspectral remote sensing image data sets. For the spectral data set of typical surface features, as there are only 25 instances in this data set, the supervised classifiers were not suitable, and thus fuzzy c-means (FCM) was used as the classifier. In addition, the correct number of classifications is used as a criterion for judging good or bad.

B. UCI Data Sets

The ten UCI data sets were obtained from the Center for Machine Learning and Intelligent Systems at the UCI, Irvine, CA, USA. The names of the data sets are Balance scale, Wine, Zoo, Waveform Version 1, Spambase, Sonar, Musk Version 2, Msplice, Madelon, and Isolet5. The details (the number of instances, features, and classes; and the date) of the selected UCI data sets are presented in Table I.

We started by selecting a subset that contained two features and tested it to a certain size. In addition, according to the number of features of the different UCI data sets, the number of times that the experiments were repeated was dependent on the range of the size of the selected feature subset. As shown in Table I, the range for the Spambase data set is 2–55, and the ranges for the Musk Version 2, Msplice, Madelon, and Isolet5 data sets are all 2–100. The SVM and KNN classifiers were used to test the selected feature subsets, where 5% of the instances were randomly selected and used to train the classifiers, and the other 95% were used as the test samples, and the error rate was recorded. Furthermore, the training and test samples in one data set were fixed for all the comparative methods. From the results shown in Fig. 9, it can be inferred that the proposed MOSCA_FS achieves a relatively low error rate, especially for the Wine, Zoo, Spambase, Musk Version 2, Madelon, and Isolet5 data sets. In order to give a more intuitive comparison, the mean and standard deviation of the different comparative methods are listed in Table II. Clearly, the EC-based feature selection methods are superior to the traditional and the non-EC-based methods, and the evolutionary multiobjective feature selection methods achieve better performances than the single-objective based methods. Furthermore, for the SVM classifier, the MOSCA_FS method achieves a first place on the Balance, Wine, Zoo, Spambase, and Sonar data sets, and obtains a second place on all the other UCI data sets. In addition, for the KNN classifier, the proposed

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

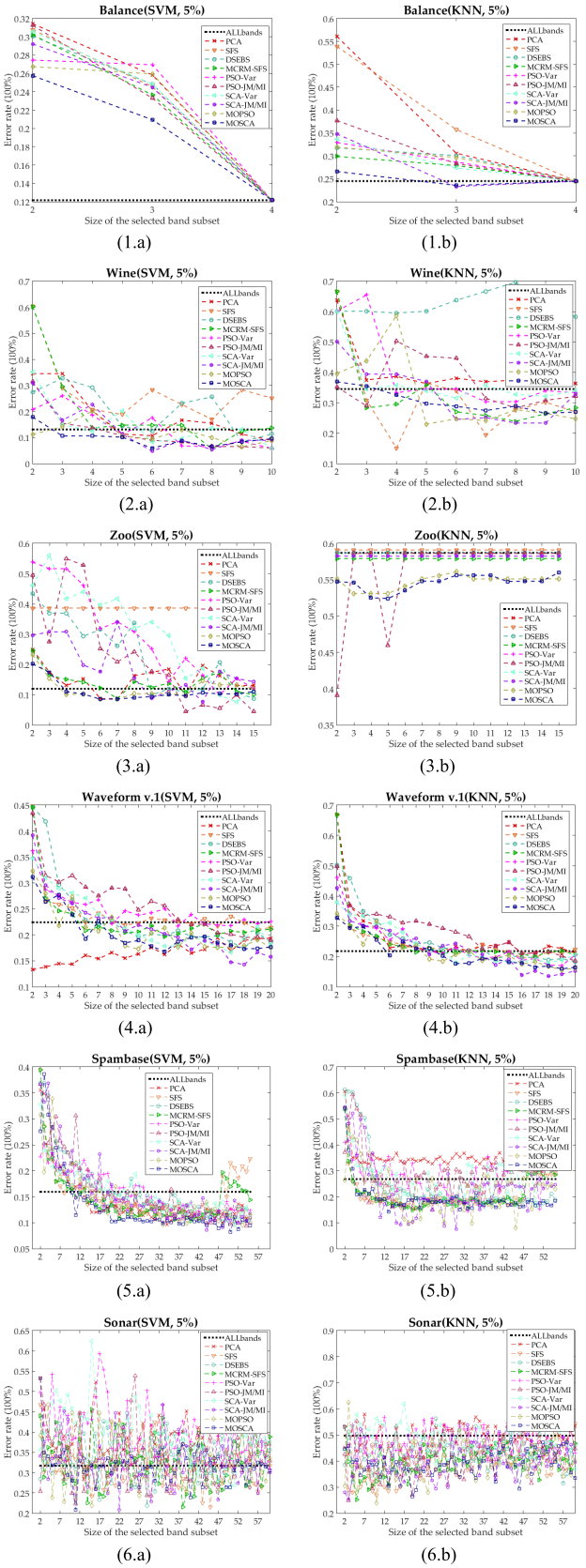


Fig. 9. Results of different methods for the ten UCI data sets. (1.a) Balance-SVM. (1.b) Balance-KNN. (2.a) Wine-SVM. (2.b) Wine-KNN. (3.a) Zoo-SVM. (3.b) Zoo-KNN. (4.a) Waveform v.1-SVM. (4.b) Waveform v.1-KNN. (5.a) Spambase-SVM. (5.b) Spambase-KNN. (6.a) Sonar-SVM.

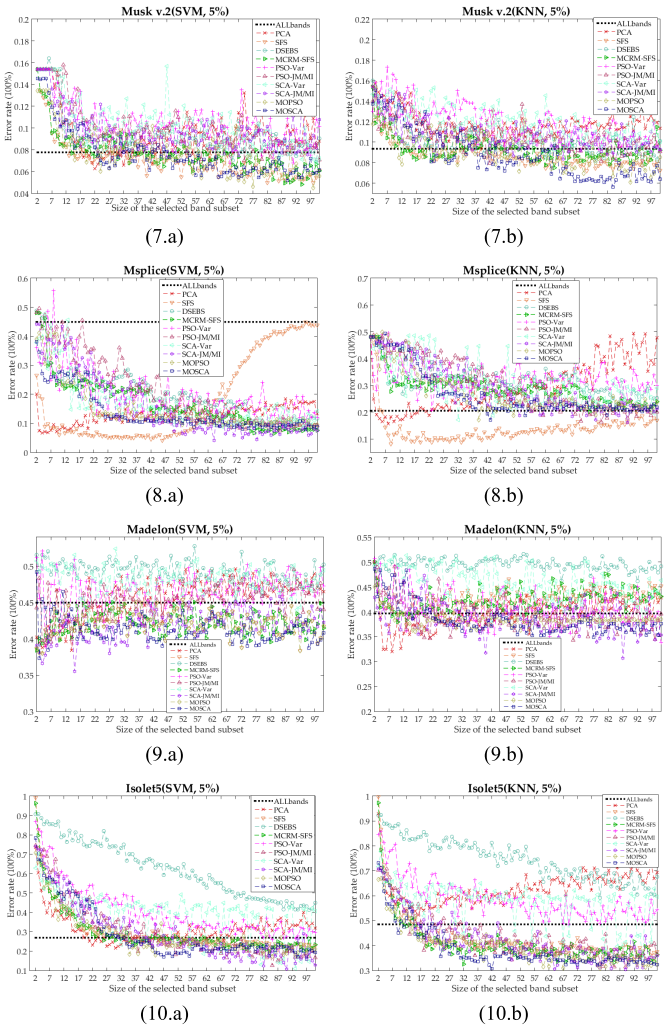


Fig. 9. *(Continued.)* Results of different methods for the ten UCI data sets. (7.a) Musk v.2-SVM. (7.b) Musk v.2-KNN. (8.a) Msplice-SVM. (8.b) Msplice-KNN. (9.a) Madelon-SVM. (9.b) Madelon-KNN. (10.a) Isolet5-SVM. (10.b) Isolet5-KNN.

method achieves a first place on the Balance, Wine, Zoo, Spambase, Sonar, and Isolet5 data sets, and second place on all the other data sets. Furthermore, the CRM-based filter method obtains an average performance due to the lack of good global and local search capabilities and the tradeoff ability between the multiple objectives, but it still performs better than the traditional PCA method and some of the EC-based feature selection methods. Overall, the proposed MOSCA_FS method achieves the best performance among all the methods.

C. Hyperspectral Remote Sensing Image Data Sets

In order to verify the universal performance of the proposed method, another type of data set was applied: hyperspectral remote sensing imagery. These images are obtained by imaging the same coverage area with a spectral imager mounted on an aviation or aerospace platform. Hyperspectral images have more feature bands than natural images, and a pixel represents an instance. Considering the more complex imaging

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

TABLE II
CLASSIFICATION PERFORMANCE FOR THE UCI DATA SETS

FS method	Classifier	Balance scale (2-4)	Wine (2-10)	Zoo (2-15)	Waveform v.1 (2-20)	Spambase (2-55)
ALL-Features	SVM	0.1216	0.1310	0.1196	0.2243	0.1594
PCA		0.2314±0.0991	0.1811±0.0847	0.1493±0.0443	**0.1670±0.0188**	**0.1431±0.0563**
SFS		0.2250±0.0948	0.2636±0.1222	0.3864±0.0000	0.2413±0.0508	0.1575±0.0498
DSEBS		0.2235±0.0921	0.1786±0.0915	0.2188±0.1270	0.2324±0.0744	0.1631±0.0619
MCRM-SFS		0.2202±0.0914	0.2045±0.0.1514	0.1389±0.0382	0.2267±0.0547	0.1479±0.0490
PSO-Var		0.2219±0.0869	0.1212±0.0698	0.2857±0.1580	0.2461±0.0346	0.1564±0.0401
PSO-JM/MI		0.2224±0.0959	0.1032±0.0753	0.2168±0.1778	0.2555±0.0619	0.1552±0.0627
SCA-Var		0.2252±0.0942	0.1389±0.0775	0.2923±0.1577	0.2213±0.0519	0.1538±0.0431
SCA-JM/MI		0.2196±0.0881	0.1151±0.0801	0.1801±0.0915	0.2134±0.0600	0.1521±0.0640
MOPSO		**0.2162±0.0820**	**0.1031±0.0406**	**0.1230±0.0374**	0.2076±0.0416	0.1451±0.0504
MOSCA_FS		**0.1962±0.0689**	**0.0989±0.0295**	**0.1113±0.0326**	**0.2033±0.0397**	**0.1315±0.0606**
ALL-Features	KNN	0.2449	0.3452	0.5870	0.2178	0.2679
PCA		0.3705±0.1676	0.3953±0.0768	0.5870±0.0000	0.2631±0.1038	0.3517±0.0414
SFS		0.3806±0.1485	0.3118±0.1285	0.5909±0.0000	0.2523±0.1009	0.2182±0.0644
DSEBS		0.2977±0.0380	0.6146±0.0379	0.5870±0.0000	0.2537±0.0896	0.2411±0.1205
MCRM-SFS		**0.2742±0.0273**	0.3087±0.1199	0.5791±0.0000	0.2493±0.1024	**0.2120±0.0674**
PSO-Var		0.2869±0.0421	0.3854±0.1161	0.5824±0.0000	0.2380±0.0704	0.2752±0.0682
PSO-JM/MI		0.3017±0.0681	0.3512±0.0737	0.5632±0.0578	0.2783±0.0766	0.2720±0.0863
SCA-Var		0.2849±0.0467	0.3566±0.0787	0.5824±0.0000	0.2354±0.0646	0.2733±0.0787
SCA-JM/MI		0.2753±0.0632	**0.3067±0.0893**	0.5824±0.0000	0.2237±0.0769	0.2343±0.1059
MOPSO		0.2869±0.0381	0.3069±0.1098	**0.5474±0.0093**	**0.2150±0.0527**	0.2273±0.0828
MOSCA_FS		**0.2488±0.0155**	**0.2952±0.0361**	**0.5464±0.0106**	**0.2101±0.0506**	**0.1928±0.0596**
FS methods	**Classifier**	**Sonar (2-60)**	**Musk v.2 (2-100)**	**Msplice (2-100)**	**Madelon (2-100)**	**Isolet5 (2-100)**
ALL-Features	SVM	0.3172	0.0777	0.4493	0.4498	0.2692
PCA		0.3937±0.0575	0.0946±0.0216	**0.1301±0.0300**	0.4567±0.0278	0.3168±0.0695
SFS		0.3208±0.0530	**0.0712±0.0180**	0.1805±0.1486	0.4209±0.0160	0.3013±0.1251
DSEBS		0.3490±0.0567	0.0937±0.0209	0.2015±0.0823	0.4931±0.0145	0.6224±0.1437
MCRM-SFS		0.3277±0.0587	0.0767±0.0176	0.1692±0.0875	0.4216±0.0157	0.2957±0.1210
PSO-Var		0.3969±0.0689	0.1032±0.0203	0.1947±0.0838	0.4652±0.0237	0.4080±0.1329
PSO-JM/MI		0.3280±0.0630	0.0986±0.0225	0.2094±0.1219	0.4522±0.0216	0.3011±0.1481
SCA-Var		0.3962±0.0653	0.1031±0.0224	0.1694±0.0907	0.4808±0.0186	0.3924±0.1159
SCA-JM/MI		0.3220±0.0592	0.0963±0.0185	0.1614±0.1141	0.4187±0.0202	0.2976±0.1471
MOPSO		**0.3191±0.0596**	0.0777±0.0202	0.1511±0.0831	**0.4035±0.0204**	**0.2837±0.1070**
MOSCA_FS		**0.3155±0.0440**	**0.0761±0.0215**	**0.1383±0.0695**	**0.4130±0.0179**	**0.2893±0.1388**
ALL-Features	KNN	0.4975	0.0935	0.2054	0.3968	0.4852
PCA		0.4849±0.0541	0.1184±0.0125	0.2929±0.0924	0.3959±0.0299	0.4932±0.0475
SFS		**0.3904±0.0678**	**0.0876±0.0125**	**0.1402±0.0618**	0.4250±0.0199	0.4300±0.0985
DSEBS		0.4341±0.0617	0.1011±0.0144	0.3154±0.0495	0.4955±0.0108	0.7475±0.0817
MCRM-SFS		0.3979±0.0536	0.0931±0.0119	0.2886±0.0559	0.4239±0.0226	0.4125±0.0949
PSO-Var		0.4741±0.0513	0.1193±0.0180	0.3270±0.0531	0.4080±0.0343	0.5891±0.0810
PSO-JM/MI		0.4187±0.0769	0.1062±0.0147	0.3023±0.0979	0.3950±0.0317	0.4344±0.0903
SCA-Var		0.4752±0.0537	0.1193±0.0158	0.3044±0.0833	0.4589±0.0335	0.5732±0.0787
SCA-JM/MI		0.4063±0.0659	0.1055±0.0147	0.2914±0.0927	0.3863±0.0294	0.4245±0.0934
MOPSO		0.4014±0.0651	0.0908±0.0162	0.2768±0.0806	**0.3846±0.0143**	**0.3977±0.0738**
MOSCA_FS		**0.3873±0.0493**	**0.0883±0.0241**	**0.2704±0.0871**	**0.3854±0.0325**	**0.3863±0.0827**

method and the inherently complex structure of hyperspectral images, they pose a greater processing challenge. Thus, the process of hyperspectral feature selection can help obtain higher classification accuracy. In this study, five hyperspectral remote sensing image data sets were utilized. The first two hyperspectral remote sensing images are public data sets.

1) The Indian Pines image data set [82].
2) The Salinas image data set [83]. The WuHan University Hyperspectral Image (WHU-Hi) data sets were obtained by an unmanned aerial vehicle (UAV) platform equipped with Headwall Nano hyperspectral imaging spectrometer [84] from the Intelligent Data Extraction and Analysis of Remote Sensing (RSIDEA) Group of Wuhan University, Wuhan, China. In this article, three WHU-Hi subdata sets are used to test the algorithms.
3) The WHU-Hi-HongHu data set.
4) The WHU-Hi-HanChuan data set.
5) The WHU-Hi-Mountain data set spliced in this study.

The five image data sets have different application scenes. The first three image data sets involve pure agricultural classification, the fourth image data set involves the mixed classification of natural land covers and crops, and the last image data set involves the classification of a mountain area. In the different tasks, the ground objects that need attention are different, and thus the selected feature bands are also different. In addition, these five hyperspectral image data sets were obtained by different sensors. The first two image data sets were collected by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS) and the last three image data sets were obtained with a Headwall Nano-Hyperspec spectrometer. Meanwhile, from the spectral curves of the data sets, it can be found that the distribution relationships between the spectral curves are

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

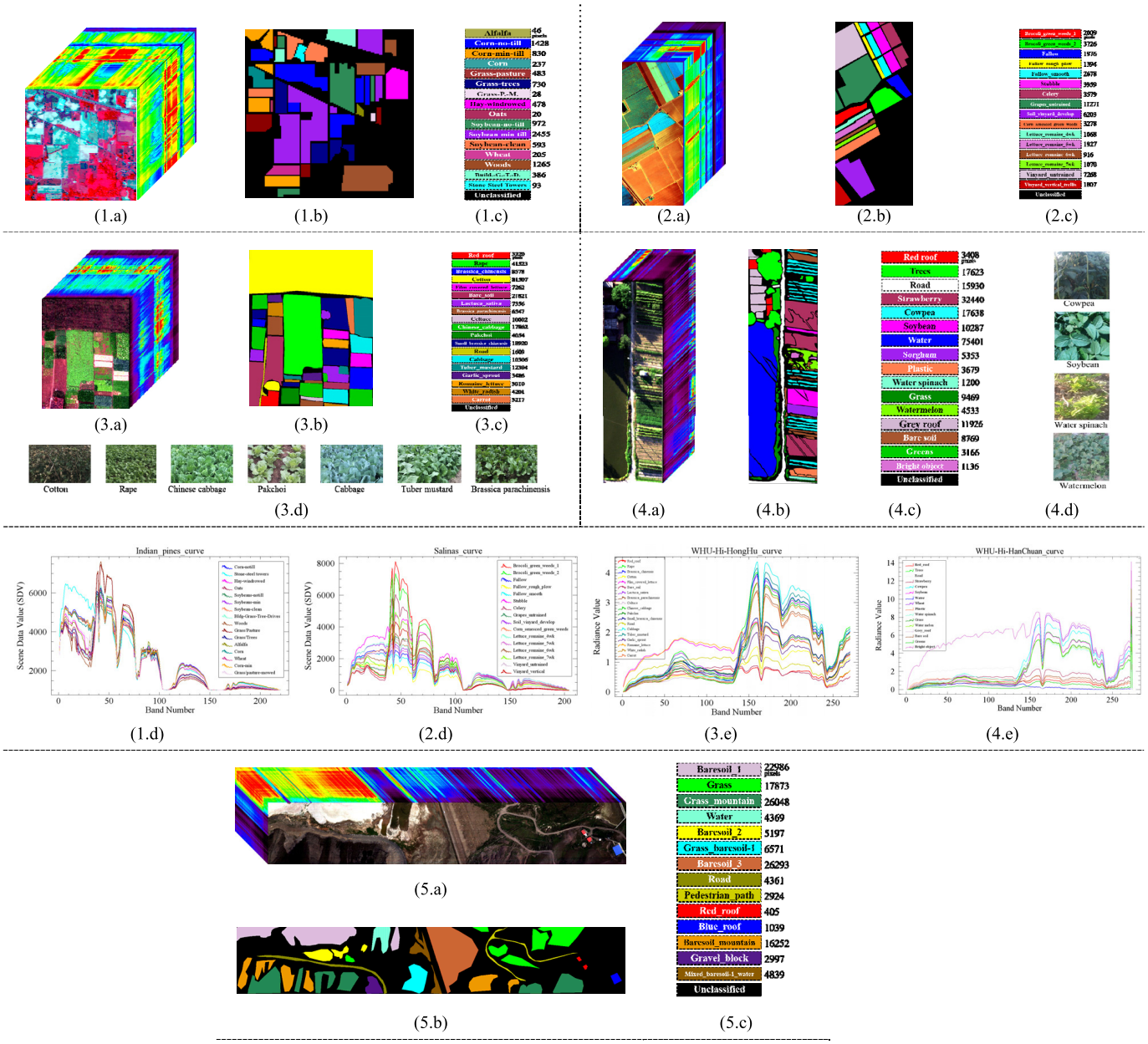


Fig. 10. Five hyperspectral remote sensing image data sets. (1) Indian Pines image data set. (2) Salinas image data set. (3) WHU-Hi-HongHu image data set. (4) WHU-Hi-HanChuan image data set. (5) WHU-Hi-Mountain image data set. (1.a) Image. (1.b) G-T data. (1.c) Legend. (1.d) Spectral curves. (2.a) Image. (2.b) G-T data. (2.c) Legend. (2.d) Spectral curves. (3.a) Image. (3.b) G-T data. (3.c) Legend. (3.d) Some typical crop species in the study area. (3.e) Spectral curves. (4.a) Image. (4.b) G-T data. (4.c) Legend. (4.d) Some typical crop species in the study area. (4.e) Spectral curves. (5.a) Image. (5.b) G-T data. (5.c) Legend. (G-T: ground truth).

different, which indicates that these data sets have different intraclass and interclass differences, which will directly lead to different classification difficulties. Thus, these five hyperspectral image data sets can show the effectiveness and universality of the proposed method from different perspectives. Details of the five hyperspectral image data sets are given in Fig. 10 and Table III.

From the descriptions of the five image data sets, the crucial information is as follows: 1) the number of features is large, which means that the dimension of the feature space is ultrahigh, making the model difficult to interpret, more likely to result in overfitting, and achieve a poor classification accuracy; 2) the number of classes for the image data sets is larger than that for the UCI data sets, which means that the situation of misclassification between categories will increase; and 3) from the spectral curves of the data sets, it can be seen that they are very close, from which it can be inferred that the situation of misclassification between categories will become worse. The results obtained in this study are given in Fig. 10, where the ranges of the size of the selected feature subsets were all set as 4–60. In addition, 10% of the labeled samples were randomly selected and used to train the classifier, and the other 90% were used as the test samples. Furthermore, the training and test samples in one data set were fixed for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

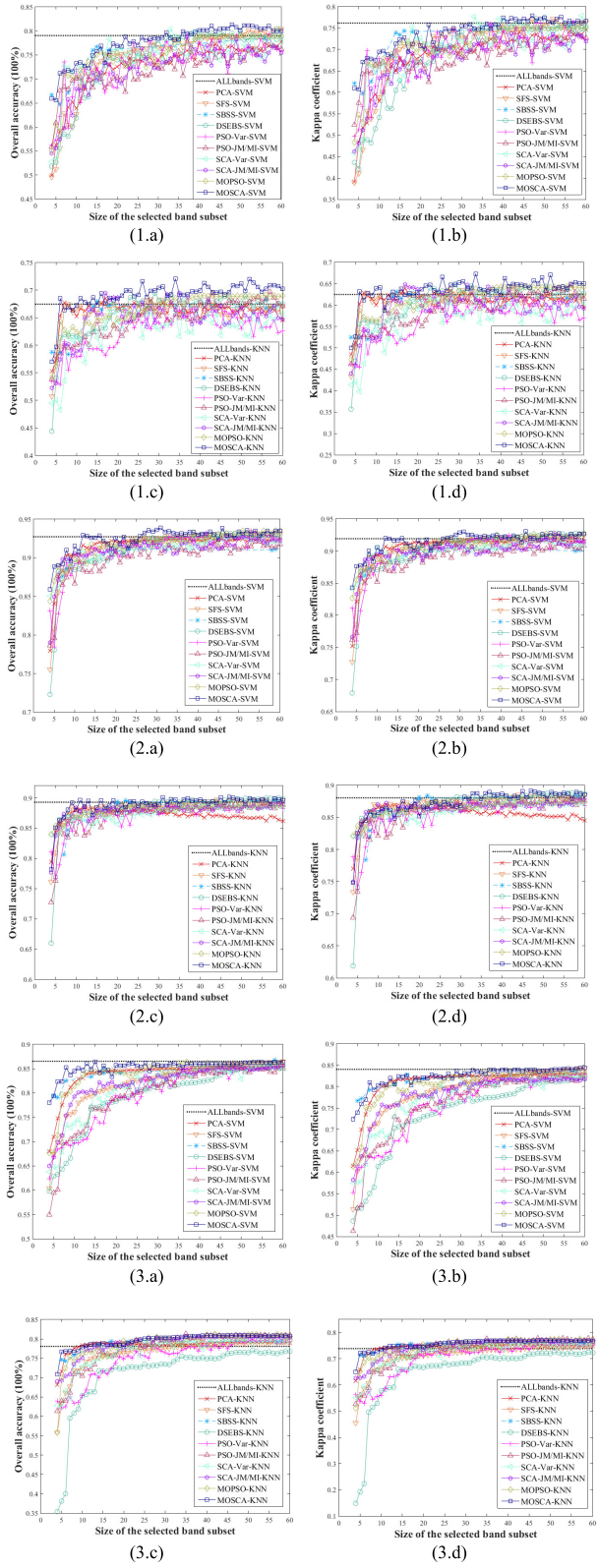


Fig. 11. Results of different methods for the five hyperspectral remote sensing image data sets. (1) Indian Pines image data set. (2) Salinas image data set. (3) WHU-Hi-HongHu image data set. (4) WHU-Hi-HanChuan image data set. (5) WHU-Hi-Mountain image data set. (1.a) Indian-SVM-OA. (1.b) Indian-SVM-Kappa. (1.c) Indian-KNN-OA. (1.d) Indian-KNN-Kappa. (2.a) Salinas-SVM-OA. (2.b) Salinas-SVM-Kappa. (2.c) Salinas-KNN-OA. (2.d) Salinas-KNN-Kappa. (3.a) WHU-Hi-HH-SVM-OA. (3.b) WHU-Hi-HH-SVM-Kappa. (3.c) WHU-Hi-HH-KNN-OA. (3.d) WHU-Hi-HH-KNN-Kappa.

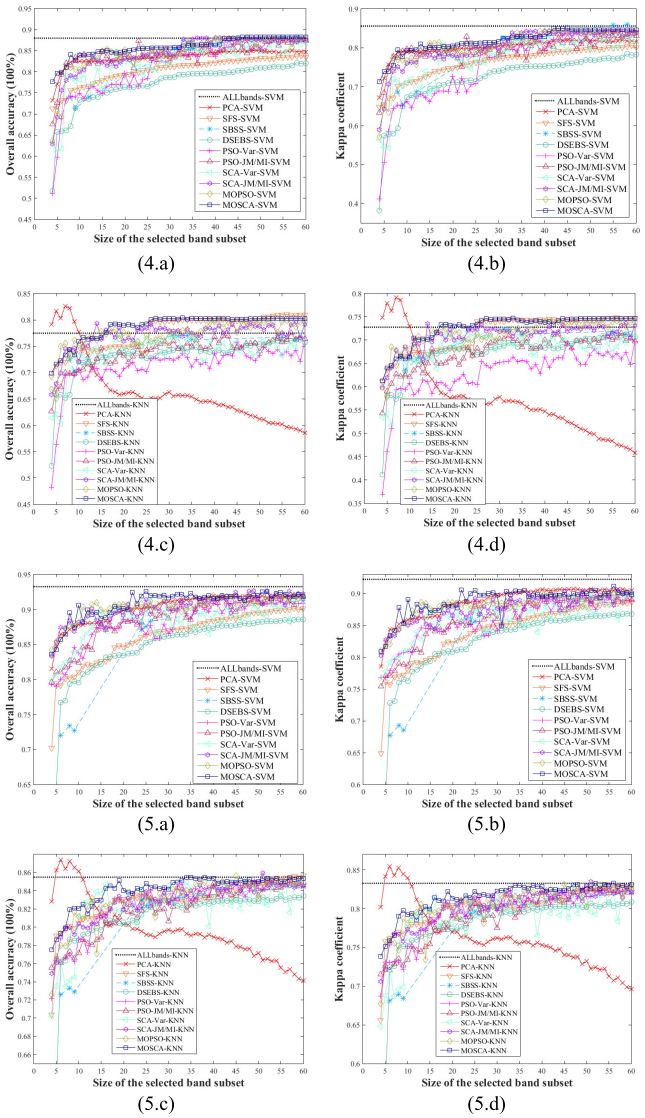


Fig. 11. *(Continued.)* Results of different methods for the five hyperspectral remote sensing image data sets. (1) Indian Pines image data set. (2) Salinas image data set. (3) WHU-Hi-HongHu image data set. (4) WHU-Hi-HanChuan image data set. (5) WHU-Hi-Mountain image data set. (4.a) WHU-Hi-HC-SVM-OA. (4.b) WHU-Hi-HC-SVM-Kappa. (4.c) WHU-Hi-HC-KNN-OA. (4.d) WHU-Hi-HC-KNN-Kappa. (5.a) WHU-Hi-Mountain-SVM-OA. (5.b) WHU-Hi-Mountain-SVM-K. (5.c) WHU-Hi-Mountain-KNN-OA. (5.d) WHU-Hi-Mountain-KNN-K.

all the comparative methods. The OA and Kappa are used as the judgment indices.

From the results of the different feature selection methods, as shown in Fig. 11, it can be found that the proposed method—MOSCA_FS—achieves relatively high values of OA and Kappa with both classifiers. In particular, MOSCA_FS can achieve a better performance when the size of the selected feature subset is small. In order to give a more intuitive comparison, the mean and standard deviation values of the different methods are listed in Tables IV–VIII. In addition, the optimal classification OA and Kappa values are shown in Tables IX and X.

Different from the relatively simple UCI data sets, the hyperspectral image data sets have more classes and feature

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

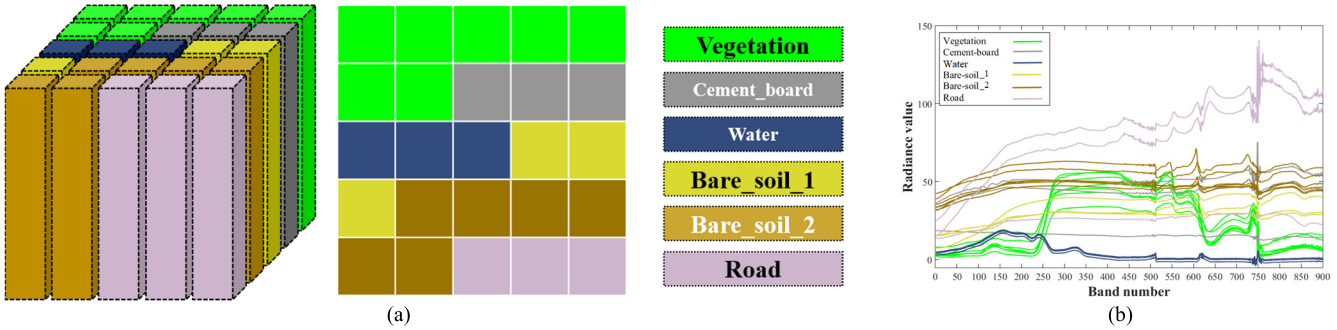


Fig. 12. (a) Spectral data set of typical surface features. (b) Spectral curves.

TABLE III
DETAILS OF THE FIVE HYPERSPECTRAL IMAGE DATA SETS

Attributes	1. Ind.	2. Sa.	3. HH	4. HC	5. Mou.
# Reso.	20 m	3.7 m	0.04 m	0.1 m	0.56 m
# Features	220	204	270	274	270
# Classes	16	16	19	16	14
# Size	145×145	217×512	475×600	303×1050	1460×250
# True c.	26-14-8	26-14-8	108-68-27	110-70-29	108-68-27

Note: Reso.: spatial resolution of the images; True c.: true color, the R-G-B bands of the images; Ind: Indian Pines, Sa: Salinas, HH: WHU-Hi-HongHu, HC: WHU-Hi-HanChuan, and Mou.: WHU-Hi-Mountain.

TABLE IV
CLASSIFICATION PERFORMANCE FOR THE INDIAN PINES HYPERSPECTRAL IMAGE DATA SET (THE AVERAGE OA ± THE STANDARD DEVIATION)

	Classifier: SVM		Classifier: KNN	
	A_OA±Std	A_K±Std	A_OA±Std	A_K±Std
1	0.7905	0.7615	0.6747	0.6249
2	0.7262±0.057	0.6846±0.070	0.6632±0.021	0.6127±0.024
3	0.7464±0.072	0.7065±0.085	0.6604±0.027	0.6106±0.031
4	0.7478±0.064	**0.7120±0.074**	0.6441±0.044	0.5906±0.051
5	0.7400±0.072	0.6864±0.072	0.6588±0.041	0.6078±0.048
6	0.7323±0.051	0.6938±0.059	0.6256±0.034	0.5664±0.040
7	0.7203±0.039	0.6830±0.039	0.6473±0.034	0.5902±0.041
8	0.7431±0.055	0.7012±0.066	0.6284±0.041	0.5704±0.050
9	0.7209±0.053	0.6796±0.063	0.6494±0.036	0.5942±0.042
10	**0.7519±0.049**	0.7105±0.054	**0.6650±0.031**	**0.6153±0.035**
11	**0.7760±0.037**	**0.7363±0.040**	**0.6911±0.025**	**0.6355±0.028**

Note: 1: Allbands, 2: PCA, 3: SFS, 4: SBSS, 5: DSEBS, 6: PSO-Var, 7: PSO-JM/MI; 8: SCA-Var, 9: SCA-JM/MI, 10: MOPSO, 11: MOSCA_FS.

bands with closer spectral features, and thus they are more challenging. From Tables IV–VIII, for the SVM classifier, the proposed method achieves the best performance in average OA and Kappa for the Indian Pines, Salinas, WHU-Hi-HongHu, WHU-Hi-HanChuan, and WHU-Hi-Mountain data sets. For the KNN classifier, MOSCA_FS also obtains the best results among all the hyperspectral remote sensing image data sets and all the comparative methods. Furthermore, for Indian Pines, WHU-Hi-HongHu and WHU-Hi-Hanchuan image data sets, the average OA and Kappa of the proposed method are higher than those obtained using all the features in the results of the KNN classifier. Thus, the experimental results confirm that MOSCA_FS achieves superior performance in the preservation of the crucial information and the removal of redundant information for hyperspectral image classification. Of course, we also need to pay attention to the classification

TABLE V
CLASSIFICATION PERFORMANCE FOR THE SALINAS HYPERSPECTRAL IMAGE DATA SET (THE AVERAGE OA ± THE STANDARD DEVIATION)

	Classifier: SVM		Classifier: KNN	
	A_OA±Std	A_K±Std	A_OA±Std	A_K±Std
1	0.9273	0.9191	0.8928	0.8804
2	0.9154±0.023	0.9061±0.026	0.8711±0.013	0.8561±0.014
3	0.9156±0.030	0.9058±0.034	**0.8844±0.022**	**0.8710±0.024**
4	0.9009±0.014	0.8905±0.017	0.8726±0.033	0.8578±0.037
5	0.9111±0.035	0.9009±0.041	0.8820±0.035	0.8683±0.040
6	0.9105±0.027	0.9003±0.031	0.8714±0.024	0.8564±0.027
7	0.9003±0.025	0.8890±0.028	0.8719±0.029	0.8570±0.032
8	0.9104±0.021	0.8986±0.023	0.8749±0.013	0.8610±0.013
9	0.9110±0.017	0.9010±0.016	0.8782±0.017	0.8641±0.019
10	**0.9166±0.018**	**0.9071±0.020**	0.8796±0.010	0.8655±0.012
11	**0.9247±0.015**	**0.9149±0.016**	**0.8887±0.017**	**0.8721±0.022**

TABLE VI
CLASSIFICATION PERFORMANCE FOR THE WHU-HI-HONGHU HYPERSPECTRAL IMAGE DATA SET (THE AVERAGE OA ± THE STANDARD DEVIATION)

	Classifier: SVM		Classifier: KNN	
	A_OA±Std	A_K±Std	A_OA±Std	A_K±Std
1	0.8653	0.8404	0.7808	0.7389
2	0.8379±0.034	**0.8109±0.041**	0.7840±0.018	0.7433±0.022
3	0.8210±0.051	0.7884±0.064	0.7641±0.042	0.7134±0.053
4	0.8334±0.026	0.8068±0.025	0.7765±0.021	0.7424±0.020
5	0.7932±0.066	0.7367±0.088	0.7137±0.090	0.6532±0.123
6	0.7970±0.068	0.7592±0.082	0.7507±0.051	0.7026±0.067
7	0.7985±0.063	0.7582±0.076	0.7766±0.048	0.7320±0.060
8	0.8094±0.056	0.7731±0.066	0.7696±0.041	0.7246±0.050
9	0.8206±0.045	0.7829±0.053	0.7872±0.029	**0.7449±0.037**
10	**0.8395±0.039**	0.8053±0.046	**0.7852±0.039**	0.7441±0.041
11	**0.8531±0.016**	**0.8236±0.025**	**0.7966±0.018**	**0.7565±0.020**

performance of each method. From Tables IX and X, we can find that the proposed method—MOSCA_FS—obtains the highest OA and Kappa in most cases. Overall, from the experiments with hyperspectral remote sensing image data sets, the superior capability of the proposed method is again apparent.

D. Spectral Data Set of Typical Surface Features

Different from the two former types of data set, the spectral data set of the typical surface features has more feature bands, which results in a huge search space and a greater challenge to the proposed method. Thus, in order to further

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

TABLE VII

CLASSIFICATION PERFORMANCE FOR THE WHU-HI-HANCHUAN HYPERSPECTRAL IMAGE DATA SET (THE AVERAGE OA ± THE STANDARD DEVIATION)

	Classifier: SVM		Classifier: KNN	
	A_OA±Std	A_K±Std	A_OA±Std	A_K±Std
1	0.8798	0.8558	0.7749	0.7278
2	0.8346±0.020	0.8010±0.024	0.6642±0.063	0.5765±0.086
3	0.8037±0.032	0.7622±0.040	**0.7772±0.032**	**0.7156±0.038**
4	0.8192±0.068	0.7824±0.082	0.7506±0.026	0.6977±0.034
5	0.7710±0.053	0.7202±0.069	0.7311±0.038	0.6739±0.048
6	0.8109±0.071	0.7553±0.093	0.7122±0.043	0.6298±0.054
7	0.8419±0.034	0.7962±0.034	0.7390±0.028	0.6753±0.035
8	0.8220±0.058	0.7703±0.067	0.7396±0.035	0.6807±0.042
9	0.8428±0.047	0.8019±0.049	0.7706±0.027	0.7091±0.028
10	**0.8444±0.046**	**0.8042±0.051**	0.7731±0.026	0.7142±0.032
11	**0.8586±0.023**	**0.8183±0.030**	**0.7878±0.025**	**0.7265±0.032**

TABLE VIII

CLASSIFICATION PERFORMANCE FOR THE WHU-HI-MOUNTAIN HYPERSPECTRAL IMAGE DATA SET (THE AVERAGE OA ± THE STANDARD DEVIATION)

	Classifier: SVM		Classifier: KNN	
	A_OA±Std	A_K±Std	A_OA±Std	A_K±Std
1	0.9326	0.9224	0.8550	0.8327
2	0.9022±0.022	0.8873±0.025	0.7970±0.033	0.7670±0.039
3	0.8612±0.039	0.8394±0.046	0.8288±0.024	0.8010±0.027
4	0.8346±0.082	0.8093±0.095	0.7976±0.052	0.7653±0.061
5	0.8467±0.050	0.8223±0.060	0.8099±0.041	0.7798±0.049
6	0.8812±0.033	0.8583±0.034	0.8253±0.029	0.7970±0.032
7	0.8865±0.033	0.8597±0.039	0.8193±0.027	0.7909±0.033
8	0.8829±0.027	0.8579±0.030	0.8223±0.032	0.7883±0.037
9	0.9013±0.019	0.8761±0.022	0.8281±0.022	0.8003±0.025
10	**0.9032±0.016**	**0.8809±0.020**	**0.8305±0.025**	**0.8037±0.029**
11	**0.9084±0.020**	**0.8878±0.022**	**0.8410±0.018**	**0.8147±0.021**

confirm the capability of the proposed method, the spectral data set of the typical surface features of a mining area was also utilized in the experiments. This data set was obtained via an SVC HR-1024 portable spectrometer produced by the Spectra Vista Corporation of America, Poughkeepsie, NY, USA. A description of the data set is given in Fig. 12. In this data set, there are 25 instances, and six typical spectral features: vegetation, cement_board, water, bare_soil_1, bare_soil_2, and road. Furthermore, there are 1024 feature bands in the original data, and 900 feature bands were applied in the experiments after removing the noisy bands. For the subsequent experiments, after the different feature selection methods were applied, the FCM algorithm was used to cluster the selected features into six classes, and the number of correct classifications was counted. Feature subset sizes of 50, 100, and 200 were tested for the subsequent clustering. The results for different comparison methods are presented in Table XI.

As shown in Table XI, it can be found that when the size of the feature subset is 50, the numbers of correct classifications for all the methods are relatively low, and when the size of the feature subset reaches 100, the numbers of correct classifications increase for all the methods. Furthermore, when the size of the feature subset is 200, the numbers tend to remain stable. In addition, the proposed method—MOSCA_FS—achieves the highest number of correct classifications among all the comparative methods in all three sizes of feature subset. Thus,

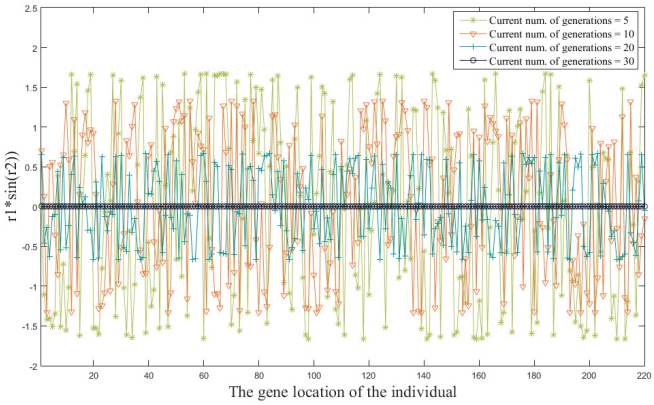


Fig. 13. Parameter analysis for the "$r_1 \times \sin(r_2)$."

for the spectral data set of typical surface features, which has a large number of feature bands, the proposed method can again obtain superior performance. Overall, through the experiments with this third type of data set, the superior capability and universality of the proposed method is again confirmed.

E. Computational Complexity Analysis

For the computational complexity of the proposed method, we assume that the number of features of the original data is L, the number of instances is $\Im = M \times N$, and the maximum number of iterations is $T_{\max}$. Thus, the computational complexity of the MOSCA_FS method can be split into three main steps.

1) *Initialization of the Solutions and the Population*: There are NP solutions for the hyperspectral feature selection in a population, and the number of variables in one solution is L. Thus, the computational complexity of this step is $O(\mathrm{NP} \times L)$.
2) *Calculation of the Values of the Objective Functions and the* Selection *of the Destination Point*:
   a) Assuming that L' features are selected from the original features L, the computational complexity for calculation of the value of the function f_1 is $\hbar = O(\mathrm{NP} \times L' \times \Im)$, and that for the function f_2 is $\lambda = O(\mathrm{NP} \times C_{L'}^2 \times \Im^2)$.
   b) The computational complexity for the selection of the destination point is $O(\mathrm{NP})$.
3) *The Main Loop of the Evolutionary Feature Selection:* There are $T_{\max}$ iterations of evolutionary operation for hyperspectral feature selection.
   a) The computational complexity for evolutionary operation by discrete SCA is $O(T_{\max} \times \mathrm{NP} \times L)$.
   b) After executing the entire loop, the computational complexity for calculation of the value of the function f_1 is $O(T_{\max} \times \hbar)$, and that for the function f_2 is $O(T_{\max} \times \lambda)$. The computational complexity of updating the destination point is $O(T_{\max} \times \mathrm{NP})$.

F. Sensitivity Analysis for the Parameters of the Proposed Method

1) *Analysis for Parameter "$r_1 \times sin\,(r_2)$":* Taking the Indian Pines hyperspectral remote sensing image data set as an example, we recorded the values of $r_1 \times \sin\,(r_2)$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
OPTIMAL CLASSIFICATION OA AND KAPPA COEFFICIENT FOR EACH METHOD (SVM CLASSIFIER)

Methods	Indian Pines		Salinas		WHU-Hi-HongHu		WHU-Hi-HanChuan		WHU-Hi-Mountain	
	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa
ALLbands-SVM	79.05%	0.7615	92.73%	0.9191	**86.53%**	0.8404	87.98%	**0.8558**	**93.26%**	**0.9224**
PCA-SVM	77.82%	0.7465	92.57%	0.9171	85.42%	0.8307	85.05%	0.8203	91.93%	0.9071
SFS-SVM	80.50%	0.7643	93.26%	0.9248	86.50%	**0.8430**	83.86%	0.8057	90.17%	0.8867
SBSS-SVM	78.73%	0.7565	91.16%	0.9052	**86.76%**	0.8339	**88.25%**	**0.8591**	91.26%	0.8894
DSEBS-SVM	79.94%	0.7755	93.20%	0.9240	85.73%	0.8204	82.91%	0.7846	88.61%	0.8686
PSO-Var-SVM	79.46%	0.7651	93.51%	**0.9277**	85.59%	0.8274	87.87%	0.8476	91.60%	0.8968
PSO-JM/MI-SVM	77.39%	0.7405	92.10%	0.9120	85.87%	0.8327	87.72%	0.8348	92.19%	0.9018
SCA-Var-SVM	**80.65%**	**0.7790**	92.88%	0.9178	85.59%	0.8274	87.33%	0.8310	91.94%	0.9025
SCA-JM/MI-SVM	78.77%	0.7590	92.83%	0.9201	85.59%	0.8182	88.01%	0.8422	92.65%	0.9049
MOPSO-SVM	78.91%	0.7500	**93.53%**	0.9267	86.47%	0.8370	87.92%	0.8423	92.06%	0.9037
MOSCA_FS-SVM	**81.32%**	**0.7821**	**93.81%**	**0.9297**	86.41%	**0.8422**	**88.27%**	0.8505	**92.56%**	**0.9118**

TABLE X
OPTIMAL CLASSIFICATION OA AND KAPPA COEFFICIENT FOR EACH METHOD (KNN CLASSIFIER)

Methods	Indian Pines		Salinas		WHU-Hi-HongHu		WHU-Hi-HanChuan		WHU-Hi-Mountain	
	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa	OA	Kappa
ALLbands-KNN	67.47%	0.6249	89.28%	0.8804	78.08%	0.7389	77.49%	0.7278	85.50%	0.8327
PCA-KNN	68.52%	0.6390	88.31%	0.8695	79.24%	0.7514	**82.62%**	**0.7911**	**87.41%**	**0.8547**
SFS-KNN	68.33%	0.6370	89.71%	0.8851	79.50%	0.7513	**81.05%**	**0.7488**	85.67%	0.8297
SBSS-KNN	68.40%	0.6369	89.59%	0.8838	79.36%	0.7573	77.13%	0.7241	84.46%	0.8203
DSEBS-KNN	69.48%	**0.6490**	**89.99%**	**0.8883**	76.96%	0.7261	76.27%	0.7129	83.46%	0.8088
PSO-Var-KNN	66.60%	0.6134	89.02%	0.8775	79.79%	0.7595	76.06%	0.6978	85.06%	0.8259
PSO-JM/MI-KNN	69.73%	0.6465	89.12%	0.8786	**81.28%**	**0.7766**	77.53%	0.7147	85.13%	0.8288
SCA-Var-KNN	67.76%	0.6284	88.64%	0.8705	79.79%	0.7595	77.58%	0.7179	85.44%	0.8281
SCA-JM/MI-KNN	**69.84%**	0.6482	89.01%	0.8775	80.65%	0.7690	79.64%	0.7362	**85.96%**	0.8352
MOPSO-KNN	69.13%	0.6437	89.03%	0.8776	**81.26%**	**0.7740**	79.86%	0.7411	85.61%	0.8314
MOSCA_FS-KNN	**72.10%**	**0.6730**	**90.13%**	**0.8916**	81.02%	0.7710	80.40%	0.7470	85.50%	**0.8333**

TABLE XI
CORRECT NUMBER OF CLASSIFICATIONS FOR EACH METHOD (FCM CLASSIFIER)

Method	PCA	SFS	DSEBS	PSO-Var	PSO-JM/MI	SCA-Var	SCA-JM/MI	MOPSO	MOSCA_FS
Correct / All (50)	17 / 25	19 / 25	**20 / 25**	19 / 25	19 / 25	17 / 25	18 / 25	19 / 25	**20 / 25**
Correct / All (100)	20 / 25	20 / 25	20 / 25	20 / 25	**22 / 25**	21 / 25	**22 / 25**	**22 / 25**	**23 / 25**
Correct / All (200)	20 / 25	19 / 25	20 / 25	**22 / 25**	**22 / 25**	20 / 25	21 / 25	**22 / 25**	**23 / 25**

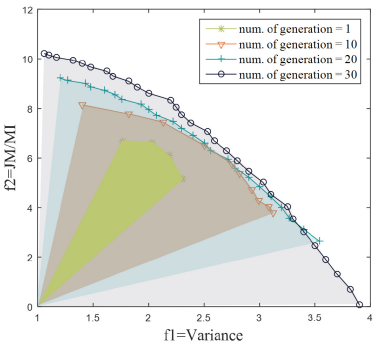


Fig. 14. Analysis of the convergence of MOSCA_FS.

for each gene location in a solution when the numbers of iterations were 5, 10, 20, and 30. The results are shown in Fig. 13, where it can be found that the values become smaller as the number of iterations increases, which means that the strategy in (6) can smoothly transition from exploration to exploitation.

2) *Analysis of the Convergence*: In this section, the Indian Pines hyperspectral remote sensing image data set is again taken as an example. When the size of the selected feature subset was 20, the convergence of the Pareto front when the numbers of iterations were 1, 10, 20, and 30 is shown in Fig. 14. It can be seen that when the number of iterations is 30, the number of solutions on the Pareto front is 30, which is equal to the preset number of solutions. Thus, the proposed method has good convergence property.

V. CONCLUSION

In this article, in order to improve the optimization capability and achieve a better balance between global exploration and local exploitation for feature selection, a novel discrete MOSCA_FS approach has been proposed. In this method, in order to achieve the retention of critical information and the removal of redundant information, an effective objective function—the ratio between the JM distance and the MI—is designed for maximizing the relevance of the selected feature subset, and the function of the Var is used to retain as much information as possible. In addition, an effective discrete SCA is designed, which uses a VSF to better capture the optimization capability, instead of just judging by a probability value,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

as in the original sigmoid function. Furthermore, in order to verify the effectiveness of the proposed method, different types of data sets were tested: ten UCI data sets, five hyperspectral image data sets, and one spectral data set of typical surface features. Through the experiments and analyses, the superior capability of the proposed method was confirmed in all the different types of data sets.

As for future work, considering that feature selection is an NP-hard problem and has a large, complex search space, a new and more powerful search method is needed. Furthermore, feature construction could be considered as a powerful strategy to improve both the prediction and classification accuracy.

REFERENCES

[1] P. Bajcsy and P. Groves, "Methodology for hyperspectral band selection," *Photogrammetric Eng. Remote Sens.*, vol. 70, no. 7, pp. 793–802, Jul. 2004.

[2] W. Sun and Q. Du, "Hyperspectral band selection: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 2, pp. 118–139, Jun. 2019.

[3] W. Shao and S. Du, "Spectral–spatial feature extraction for hyperspectral image classification: A dimension reduction and deep learning approach," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 8, pp. 4544–4554, Oct. 2016.

[4] P. Zhong, P. Zhang, and R. Wang, "Dynamic learning of SMLR for feature selection and classification of hyperspectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 5, no. 2, pp. 280–284, Apr. 2008.

[5] K. Tan, E. Li, Q. Du, and P. Du, "Hyperspectral image classification using band selection and morphological profiles," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 1, pp. 40–48, Jan. 2014.

[6] J. Li *et al.*, "Feature selection: A data perspective," *ACM Comput. Surv. (CSUR)*, vol. 50, no. 6, p. 94, Jan. 2018.

[7] P. Pudil, J. Novovičová, and J. Kittler, "Floating search methods in feature selection," *Pattern Recognit. Lett.*, vol. 15, no. 11, pp. 1119–1125, 1994.

[8] K. Z. Mao, "Orthogonal forward selection and backward elimination algorithms for feature subset selection," *IEEE Trans. Syst., Man, Cybern. B, Cybern.*, vol. 34, no. 1, pp. 629–634, Feb. 2004.

[9] J. Feng, L. C. Jiao, X. Zhang, and T. Sun, "Hyperspectral band selection based on trivariate mutual information and clonal selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 7, pp. 4092–4105, Jul. 2014.

[10] J. Im, Z. Lu, J. Rhee, and J. R. Jensen, "Fusion of feature selection and optimized immune networks for hyperspectral image classification of urban landscapes," *Geocarto Int.*, vol. 27, no. 5, pp. 373–393, Jan. 2012.

[11] M. Mahrooghy, N. H. Younan, V. G. Anantharaj, J. Aanstoos, and S. Yarahmadian, "On the use of the genetic algorithm filter-based feature selection technique for satellite precipitation estimation," *IEEE Geosci. Remote Sens. Lett.*, vol. 9, no. 5, pp. 963–967, Sep. 2012.

[12] H. Yang, Q. Du, and G. Chen, "'Particle swarm optimization-based hyperspectral dimensionality reduction for urban land cover classification," *IEEE J. Sel. Topics App. Earth Observ. Remote Sens.*, vol. 5, no. 2, pp. 544–554, Apr. 2012.

[13] H. Yang, Q. Du, H. Su, and Y. Sheng, "An efficient method for supervised hyperspectral band selection," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 1, pp. 138–142, Jan. 2011.

[14] B. Duval, J. K. Hao, and J. C. H. Hernandez, "A memetic algorithm for gene selection and molecular classification of cancer," in *Proc. 11th Annu. Conf. Genetic Evol. Comput.*, Jul. 2009, pp. 201–208.

[15] J. C. H. Hernandez, B. Duval, and J. K. Hao, "A genetic embedded approach for gene selection and classification of microarray data," in *Proc. Eur. Conf. Evol. Comput., Mach. Learn. Data Mining Bioinf.*, Apr. 2007, pp. 90–101.

[16] J. B. Lee, A. S. Woodyatt, and M. Berman, "Enhancement of high spectral resolution remote-sensing data by a noise-adjusted principal components transform," *IEEE Trans. Geosci. Remote Sens.*, vol. 28, no. 3, pp. 295–304, May 1990.

[17] C.-I. Chang, Q. Du, T.-L. Sun, and M. L. G. Althouse, "A joint band prioritization and band-decorrelation approach to band selection for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 37, no. 6, pp. 2631–2641, Nov. 1999.

[18] R. E. Roger, "A faster way to compute the noise-adjusted principal components transform matrix," *IEEE Trans. Geosci. Remote Sens.*, vol. 32, no. 6, pp. 1194–1196, Nov. 1994.

[19] W. Sun and Q. Du, "Graph-regularized fast and robust principal component analysis for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 6, pp. 3185–3195, Jun. 2018.

[20] B. Zhao, L. Gao, W. Liao, and B. Zhang, "A new kernel method for hyperspectral image feature extraction," *Geo-Spatial Info. Sci.*, vol. 20, no. 4, 309-318, 2017.

[21] S. Jia *et al.*, "3-D Gaussian–Gabor feature extraction and selection for hyperspectral imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 11, pp. 8813–8826, Nov. 2019, doi: 10.1109/TGRS.2019.2923213.

[22] Y. Yuan, X. Zheng, and X. Lu, "Discovering diverse subset for unsupervised hyperspectral band selection," *IEEE Trans. Image Process.*, vol. 26, no. 1, pp. 51–64, Jan. 2017.

[23] Y. Yuan, J. Lin, and Q. Wang, "Dual-clustering-based hyperspectral band selection by contextual analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1431–1445, Mar. 2016.

[24] Q. Wang, F. Zhang, and X. Li, "Optimal clustering framework for hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 1–13, Oct. 2018.

[25] M. Bevilacqua and Y. Berthoumieu, "Multiple-feature kernel-based probabilistic clustering for unsupervised band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6675–6689, Sep. 2019.

[26] A. Martínez-Usó, F. Pla, J. M. Sotoca, and P. García-Sevilla, "Clustering-based hyperspectral band selection using information measures," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4158–4171, Dec. 2007.

[27] Y. Liu, F. Tang, and Z. Zeng, "Feature selection based on dependency margin," *IEEE Trans. Cybern.*, vol. 45, no. 6, pp. 1209–1221, Jun. 2015.

[28] S. Gu, R. Cheng, and Y. Jin, "Feature selection for high-dimensional classification using a competitive swarm optimizer," *Soft Comput.*, vol. 22, no. 3, pp. 811–822, 2018.

[29] C. Lai, M. J. T. Reinders, and L. Wessels, "Random subspace method for multivariate feature selection," *Pattern Recognit. Lett.*, vol. 27, no. 10, pp. 1067–1076, 2006.

[30] Y. Liu, G. Wang, H. Chen, H. Dong, X. Zhu, and S. Wang, "An improved particle swarm optimization for feature selection," *J. Bionic Eng.*, vol. 8, no. 2, pp. 191–200, Jun. 2011.

[31] A. Unler and A. Murat, "A discrete particle swarm optimization method for feature selection in binary classification problems," *Eur. J. Oper. Res.*, vol. 206, no. 3, pp. 528–539, 2010.

[32] B. Xue, M. Zhang, W. N. Browne, and X. Yao, "A survey on evolutionary computation approaches to feature selection," *IEEE Trans. Evol. Comput.*, vol. 20, no. 4, pp. 606–626, Aug. 2016.

[33] W. Siedlecki and J. Sklansky, "A note on genetic algorithms for large-scale feature selection," *Pattern Recognit. Lett.*, vol. 10, no. 5, pp. 335–347, Nov. 1989.

[34] Y. S. Jeong, K. S. Shin, and M. K. Jeong, "An evolutionary algorithm with the partial sequential forward floating search mutation for large-scale feature selection problems," *J. Oper. Res. Soc.*, vol. 66, no. 4, pp. 529–538, Dec. 2015.

[35] S. Oreski and G. Oreski, "Genetic algorithm-based heuristic for feature selection in credit risk assessment," *Expert Syst. Appl.*, vol. 41, no. 4, pp. 2052–2064, 2014.

[36] C. De Stefano, F. Fontanella, C. Marrocco, and A. S. di Freca, "A GA-based feature selection approach with an application to handwritten character recognition," *Pattern Recognit. Lett.*, vol. 35, no. 1, pp. 130–141, Jan. 2014.

[37] J. H. Seo, Y. H. Lee, and Y. H. Kim, "Feature selection for very short-term heavy rainfall prediction using evolutionary computation," *Adv. Meteorol.*, vol. 2014, pp. 1–15, Jan. 2014, Art. no. 203545.

[38] E. Hancer, B. Xue, and M. Zhang, "Differential evolution for filter feature selection based on information theory and feature ranking," *Knowl.-Based Syst.*, vol. 140, pp. 103–119, Jan. 2018.

[39] L. Zhang, Y. Zhong, B. Huang, J. Gong, and P. Li, "Dimensionality reduction based on clonal selection for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 12, pp. 4172–4186, Dec. 2007.

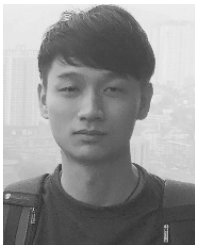
[40] Y. Zhong and L. Zhang, "A fast clonal selection algorithm for feature selection in hyperspectral imagery," *Geo-Spatial Info. Sci.*, vol. 12, no. 3, pp. 172–181, Aug. 2009.

[41] F. Xie, F. Li, C. Lei, J. Yang, and Y. Zhang, "Unsupervised band selection based on artificial bee colony algorithm for hyperspectral image classification," *Appl. Soft Comput.*, vol. 75, pp. 428–440, Feb. 2019.

[42] A. I. Hafez, H. M. Zawbaa, E. Emary, and A. E. Hassanien, "Sine cosine optimization algorithm for feature selection," in *Proc. Inter. Symp. Innov. Intell. Syst. Appl. (INISTA)*, Aug. 2016, pp. 1–5.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.

[43] M. Zhang, M. Gong, and Y. Chan, "Hyperspectral band selection based on multi-objective optimization with high information and low redundancy," *Appl. Soft Comput.*, vol. 70, pp. 604–621, Sep. 2018.
[44] Z. Wang, M. Li, and J. Li, "A multi-objective evolutionary algorithm for feature selection based on mutual information with a new redundancy measure," *Inf. Sci.*, vol. 307, pp. 73–88, Jun. 2015.
[45] H. Xia, J. Zhuang, and D. Yu, "Multi-objective unsupervised feature selection algorithm utilizing redundancy measure and negative epsi-lon-dominance for fault diagnosis," *Neurocomputing*, vol. 146, pp. 113–124, Dec. 2014.
[46] J. Lee and D.-W. Kim, "Memetic feature selection algorithm for multi-label classification," *Inf. Sci.*, vol. 293, pp. 80–96, Feb. 2015.
[47] H. Su, B. Yong, and Q. Du, "Hyperspectral band selection using improved firefly algorithm," *IEEE Geosci. Remote Sens. Lett.*, vol. 13, no. 1, pp. 68–72, Jan. 2016.
[48] S. Fallah, R. Deo, M. Shojafar, M. Conti, and S. Shamshirband, "Computational intelligence approaches for energy load forecasting in smart energy management grids: State of the art, future challenges, and research directions," *Energies*, vol. 11, no. 3, p. 596, Mar. 2018.
[49] X. Gu, Y. Li, and J. Jia, "Feature selection for transient stability assessment based on kernelized fuzzy rough sets and memetic algorithm," *Int. J. Elect. Power Energy Syst.*, vol. 64, pp. 664–670, Jan. 2015.
[50] H. H. Inbarani, M. Bagyamathi, and A. T. Azar, "A novel hybrid feature selection method based on rough set and improved harmony search," *Neural Comput. Appl.*, vol. 26, no. 8, pp. 1859–1880, Nov. 2015.
[51] A. K. Das, S. Sengupta, and S. Bhattacharyya, "A group incremental feature selection for classification using rough set theory based genetic algorithm," *Appl. Soft Comput.*, vol. 65, pp. 400–411, Apr. 2018.
[52] M. A. El Aziz and A. E. Hassanien, "Modified cuckoo search algorithm with rough sets for feature selection," *Neural Comput. Appl.*, vol. 29, pp. 925–934, Feb. 2018.
[53] B. Xue, M. Zhang, and W. N. Browne, "Particle swarm optimization for feature selection in classification: A multi-objective approach," *IEEE Trans. Cybern.*, vol. 43, no. 6, pp. 1656–1671, Dec. 2013.
[54] Z. Fei, B. Li, S. Yang, C. Xing, H. Chen, and L. Hanzo, "A survey of multi-objective optimization in wireless sensor networks: Metrics, algorithms, and open problems," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 1, pp. 550–586, 1st Quart., 2017.
[55] S. M. Jameii, K. Faez, and M. Dehghan, "AMOF: Adaptive multi-objective optimization framework for coverage and topology control in heterogeneous wireless sensor networks," *Telecommun. Syst.*, vol. 61, no. 3, pp. 515–530, 2016.
[56] R. O. Arbex and C. B. da Cunha, "Efficient transit network design and frequencies setting multi-objective optimization by alternating objective genetic algorithm," *Transp. Res. B, Methodol.*, vol. 81, pp. 355–376, Nov. 2015.
[57] P. Dai *et al.*, "Temporal information services in large-scale vehicular networks through evolutionary multi-objective optimization," *IEEE Trans. Intell. Transp. Syst.*, vol. 20, no. 1, pp. 218–231, Jan. 2019, doi: 10.1109/TITS.2018.2803842.
[58] V. Ravi, D. Pradeepkumar, and K. Deb, "Financial time series prediction using hybrids of chaos theory, multi-layer perceptron and multi-objective evolutionary algorithms," *Swarm Evol. Comput.*, vol. 36, pp. 136–149, Oct. 2017.
[59] S. Srinivasan and T. Kamalakannan, "Multi criteria decision making in financial risk management with a multi-objective genetic algorithm," *Comput. Econ.*, vol. 52, no. 2, pp. 443–457, Aug. 2018.
[60] Y. Zhong, A. Ma, Y. S. Ong, Z. Zhu, and L. Zhang, "Computational intelligence in optical remote sensing image processing," *Appl. Soft Comput.*, vol. 64, pp. 75–93, Mar. 2018.
[61] Y. Wan, Y. Zhong, and A. Ma, "Fully automatic spectral–spatial fuzzy clustering using an adaptive multiobjective memetic algorithm for multispectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2324–2340, Apr. 2019.
[62] M. Gong, H. Li, E. Luo, J. Liu, and J. Liu, "A multiobjective cooperative coevolutionary algorithm for hyperspectral sparse unmixing," *IEEE Trans. Evol. Comput.*, vol. 21, no. 2, pp. 234–248, Apr. 2017.
[63] H. Li, J. Ma, M. Gong, Q. Jiang, and L. Jiao, "Change detection in synthetic aperture radar images based on evolutionary multiobjective optimization with ensemble learning," *Memetic Comput.*, vol. 7, no. 4, pp. 275–289, Dec. 2015.
[64] A. Ma, Y. Zhong, D. He, and L. Zhang, "Multiobjective subpixel land-cover mapping," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 1, pp. 422–435, Jan. 2018.
[65] M. Song, Y. Zhong, A. Ma, and R. Feng, "Multiobjective sparse subpixel mapping for remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 7, pp. 4490–4508, Jul. 2019.
[66] B. Pan, Z. Shi, and X. Xu, "Analysis for the weakly Pareto optimum in multiobjective-based hyperspectral band selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3729–3740, Jun. 2019, doi: 10.1109/TGRS.2018.2886853.
[67] M. Črepinšek, S. H. Liu, and M. Mernik, "Exploration and exploitation in evolutionary algorithms: A survey," *ACM Comput. Surv. (CSUR)*, vol. 45, no. 3, p. 35, Jun. 2013.
[68] S. Mirjalili, "SCA: A sine cosine algorithm for solving optimization problems," *Knowl. Based Syst.*, vol. 96, pp. 120–133, Mar. 2016.
[69] J. Wang, W. Yang, P. Du, and T. Niu, "A novel hybrid forecasting system of wind speed based on a newly developed multi-objective sine cosine algorithm," *Energy Convers. Manage.*, vol. 163, pp. 134–150, May 2018.
[70] M. A. Tawhid and V. Savsani, "Multi-objective sine-cosine algorithm (MO-SCA) for multi-objective engineering design problems," *Neural Comput. Appl.*, vol. 31, pp. 1–15, Jun. 2017.
[71] M. A. Tawhid and P. Savsani, "Discrete sine-cosine algorithm (DSCA) with local search for solving traveling salesman problem," *Arabian J. Sci. Eng.*, vol. 44, no. 4, pp. 3669–3679, Apr. 2019.
[72] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multi-objective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, pp. 182–197, Apr. 2002.
[73] T. M. Cover, J. A. Thomas. *Elements of Information Theory*. Hoboken, NJ, USA: Wiley, 2012.
[74] L. Bruzzone, F. Roli, and S. B. Serpico, "An extension of the Jeffreys-Matusita distance to multiclass cases for feature selection," *IEEE Trans. Geosci. Remote Sens.*, vol. 33, no. 6, pp. 1318–1321, Nov. 1995.
[75] J. A. Richards. *An Introduction to Remote Sensing Digital Image Analysis*. Berlin, Germany: Springer-Verlag, 1986.
[76] M. J. Islam, X. Li, and Y. Mei, "A time-varying transfer function for balancing the exploration and exploitation ability of a binary PSO," *Appl. Soft Comput.*, vol. 59, pp. 182–196, Oct. 2017.
[77] J. Branke, K. Deb, H. Dierolf, and M. Osswald, "Finding knees in multi-objective optimization," in *Proc. 8th Int. Conf. Parallel Problem Solving Nature (PPSN VIII)*, Birmingham, U.K., 2004, pp. 722–731.
[78] X. Zhang, Y. Tian, and Y. Jin, "A knee point-driven evolutionary algorithm for many-objective optimization," *IEEE Trans. Evol. Comput.*, vol. 19, no. 6, pp. 761–776, Dec. 2015.
[79] T. Fei, D. Kraus, and A. Zoubir, "Contributions to automatic target recognition systems for underwater mine classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 505–518, Jan. 2015.
[80] P. Su, D. Liu, X. Li, and Z. Liu, "A saliency-based band selection approach for hyperspectral imagery inspired by scale selection," *IEEE Trans. Geosci. Remote Sens. Lett.*, vol. 15, no. 4, pp. 572–576, Apr. 2018.
[81] G. Zhu, Y. Huang, J. Lei, Z. Bi, and F. Xu, "Unsupervised hyperspectral band selection by dominant set extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 227–239, Jan. 2016.
[82] M. F. Baumgardner, L. L. Biehl, and D. A. Landgrebe. (Sep. 2015). *220 Band AVIRIS Hyperspectral Image Data Set: June 12, 1992 Indian Pine Test Site 3*. [Online]. Available: https://purr.purdue.edu/publications/1947/1
[83] J. A. Gualtieri, S. R. Chettri, R. F. Cromp, and L. F. Johnson, "Support vector machine classifiers as applied to AVIRIS data," in *Proc. 8th JPL Airborne Geosci. Workshop*, Pasadena, CA, USA, Feb. 1999, pp. 217–227.
[84] Y. Zhong *et al.*, "Mini-UAV-borne hyperspectral remote sensing: From observation and processing to applications," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 4, pp. 46–62, Dec. 2018.



Yuting Wan received the B.S. degree from the School of Geosciences and Info-Physics, Central South University, Changsha, China, in 2017. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

His research interests include remote sensing image processing, including clustering and sparse representation, and evolutionary computation, including evolutionary multiobjective optimization.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.



Ailong Ma received the B.S. degree from the China University of Petroleum, Qingdao, China, in 2010 and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2017.

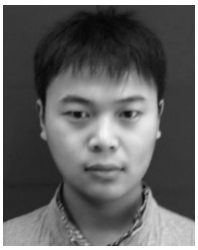
He is currently a Research Associate with Wuhan University. His major research interests are remote sensing image processing, evolutionary computing, and deep learning.



Yanfei Zhong (M'11–SM'15) received the B.S. degree in information engineering and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 2002 and 2007, respectively.

Since 2010, he has been a Full Professor with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing (LIESMARS), Wuhan University. He organized the Intelligent Data Extraction, Analysis and Applications of Remote Sensing (RSIDEA) Research Group. He has authored more than 100 research articles in international journals, such as *Remote Sensing of Environment*, the *ISPRS Journal of Photogrammetry and Remote Sensing*, and the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. His research interests include hyperspectral remote sensing information processing, high-resolution remote sensing image understanding, and geoscience interpretation for multisource remote sensing data and applications.

Dr. Zhong is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). He won the Second-Place Prize in the 2013 IEEE GRSS Data Fusion Contest and the Single-View Semantic 3-D Challenge of the 2019 IEEE GRSS Data Fusion Contest, respectively. He is currently serving as an Associate Editor for the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING and the *International Journal of Remote Sensing*.



Xin Hu received the B.S. degree in surveying and mapping engineering from Northeastern University, Shenyang, China, in 2017. He is currently pursuing the Ph.D. degree in photogrammetry and remote sensing with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China.

His major research interests include agricultural remote sensing and hyperspectral image classification.



Liangpei Zhang (M'06–SM'08–F'19) received the B.S. degree in physics from Hunan Normal University, Changsha, China, in 1982, the M.S. degree in optics from the Xi'an Institute of Optics and Precision Mechanics, Chinese Academy of Sciences, Xi'an, China, in 1988, and the Ph.D. degree in photogrammetry and remote sensing from Wuhan University, Wuhan, China, in 1998.

He was a Principal Scientist for the China State Key Basic Research Project from 2011 to 2016, appointed by the Ministry of National Science and Technology of China to lead the Remote Sensing Program in China. He is currently a "Chang-Jiang Scholar" Chair Professor appointed by the Ministry of Education of China with the State Key Laboratory of Information Engineering in Surveying, Mapping, and Remote Sensing (LIESMARS), Wuhan University. He has authored more than 700 research articles and five books. He is the Institute for Scientific Information (ISI) highly cited author. He holds 30 patents. His research interests include hyperspectral remote sensing, high-resolution remote sensing, image processing, and artificial intelligence.

Dr. Zhang is a fellow of the Institution of Engineering and Technology (IET). He was a recipient of the 2010 Best Paper Boeing Award, the 2013 Best Paper ERDAS Award from the American Society of Photogrammetry and Remote Sensing (ASPRS), and the 2016 Best Paper Theoretical Innovation Award from the International Society for Optics and Photonics (SPIE). His research teams won the top three prizes of the IEEE GRSS 2014 Data Fusion Contest, and his students have been selected as the winners or finalists of the IEEE International Geoscience and Remote Sensing Symposium (IGARSS) Student Paper Contest in recent years. He also serves as an associate editor or an editor for more than ten international journals. He is currently serving as an Associate Editor for the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING. He is the Founding Chair of the IEEE Geoscience and Remote Sensing Society (GRSS) Wuhan Chapter.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 13:05:25 UTC from IEEE Xplore. Restrictions apply.